BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Finance Panel

Kwaku Duah
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Q. Members of the Staff Finance Panel (SFP), please state your names, current employer and business address.

A. Our names are Kwaku Duah, Abdul Qadir, and Kenneth Bullock. We are employed by the New York State Department of Public Service (Department). Our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Mr. Duah, what is your position at the Department?

A. I am a Principal Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I received my Master's degree in Business Administration with a concentration in Finance and Accounting from State University of New York Institute of Technology, now known as SUNY Polytechnic Institute, in 2005.

Q. Please briefly describe your current responsibilities with the Department.

A. As a Principal Utility Financial Analyst, my assignments involve analyzing the financial condition, financing mechanisms, risk, costs of

debt and equity, diversification, and the relative competitive position of utilities and their holding company parent(s). My assignments generally involve testifying in rate cases on financial issues, analyzing financing proposals and special projects that usually involve risk and return analyses.

Q. Have you previously testified in a regulatory proceeding before the New York State Public Service Commission (Commission)?

A. Yes, I have presented testimony before the Commission in rate cases concerning NYSEG/RG&E in Cases 09-E-0715, 09-G-0716, 09-E-0717, and 09-G-0718; Niagara Mohawk in Cases 08-G-0609,10-E-0500, 12-G-0202 and 12-E-0201; Long Island Water Corporation in Case 11-W-0020; United Water New Rochelle and United Water Westchester in Cases 13-W-0539, 13-W-0564 and 14-W-0006; Central Hudson Gas and Electric in Cases 14-E-0318 and 14-G-0319; Central Hudson Gas and Electric in Cases 17-E-0459 and 17-G-00460; Suez Water New York in Cases 16-W-0130, 19-W-0168, 19-W-0269; and New York American Water Company, Inc. in Case 16-W-0259.

Q. Mr. Qadir, what is your position at the Department?

A. I am an Associate Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I graduated from the University at Albany in 2004 and received a Bachelor of Science degree with honors in Marketing and Finance. In 2007, I earned a Master's degree in Economics with a concentration in Finance and International Trade from the University at Albany. Prior to joining the Department, I worked in the banking industry, primarily as a loan officer. I joined the Department in March 2012.

Q. Please briefly describe your responsibilities with the Department.

A. My responsibilities as an Associate Utility Financial Analyst include analyzing a company's financial condition, capital structures, financing mechanisms, risks, costs of debt and equity, diversification, and the competitive position of utilities operating in New York State. My responsibilities also include

analyzing financial petitions, performing rate of return analysis and other special projects.

Q. Have you previously testified before the Commission?

Yes. I have testified in the United Water New York Inc. rate case, Case 13-W-0295; NYSEG and RG&E rate cases, Cases 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-0286; Keyspan Gas East Corporation d/b/a National Grid (KEDLI) and The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) rate cases, Cases 16-G-0058 and 16-G-0059; Corning Natural Gas Corporation (Corning) rate case, Case 16-G-0369; and Central Hudson Gas and Electric in Cases 17-E-0459 and 17-G-00460.

Q. Mr. Bullock, what is your position at the Department?

A. I am employed as a Senior Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I received a Master of Business Administration from the College of Saint Rose with an emphasis in entrepreneurship. Prior to joining the

Department, I worked in the telecommunications industry. My responsibilities in that industry included the construction and maintenance of plant, remote monitoring and diagnostics of various switching systems along with the outsourcing, design and costing of information technology networks for multi-national corporations. I joined the Department in July 2013.

Q. Please briefly describe your current responsibilities with the Department.

A. My responsibilities as a Senior Utility Financial Analyst include analyzing a company's financial condition, capital structures, financing mechanisms, risks, costs of debt and equity, diversification, and the competitive position of utilities operating in New York State.

Q. Have you previously provided testimony before the Commission?

A. Yes, I provided testimony before the Commission as a cost of capital witness in Cases 13-W-0539 and 13-W-0564, rate cases involving United Water New Rochelle and United Water Westchester; in Cases 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-

0286 rate cases involving New York State Electric & Gas Corporation and Rochester Gas and Electric Corporation; Case 16-W-0130 involving Suez Water New York; and in Case 18-G-0133 involving St. Lawrence Gas Company, Inc.

Purpose of Testimony

Q. What is the purpose of your testimony in this proceeding?

A. The purpose of our testimony is to recommend the fair rate of return to be used by the Staff Accounting Panel to determine the revenue requirement for The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) and Keyspan Gas East Corporation d/b/a National Grid (KEDLI) (collectively the Companies). Our testimony will concentrate on the appropriate ratemaking capital structure and the cost of the various capital components in the capital structure for the rate year ending March 31, 2021. We will also respond to issues raised in the direct testimonies of the Companies' capital structure witness, Mr. Jonathan Cohen and cost of equity witness, Ms. Ann E. Bulkley.

Q. Does your testimony rely on interrogatory responses (IRs) provided by the Companies?

A. Yes. These IR responses are included as Exhibit___(SFP-1).

Q. Are you sponsoring any exhibits?

A. Yes. We are sponsoring 39 exhibits as labeled below.

Exhibit Number	Description
Exhibit___(SFP-1)	Interrogatory Responses (IRs) of the Companies Supporting Staff Finance Panel Testimony;
Exhibit___(SFP-2)	KEDNY/KEDLI Capital Structure;
Exhibit___(SFP-3)	National Grid Corporate Structure;
Exhibit___(SFP-4)	New York Ring-fencing provisions;
Exhibit___(SFP-5)	Moody's Credit Opinion for KEDNY, April 8, 2019;
Exhibit___(SFP-6)	Moody's Credit Opinion for KEDLI, April 8, 2019;
Exhibit___(SFP-7)	Ms. Bulkley's Proxy Groups' Average Common Equity for 2014 to 2017
Exhibit___(SFP-8)	Standard & Poor's (S&P)

Credit Opinion on KEDNY, September 2018;
Exhibit___(SFP-9) Standard & Poor's (S&P)
Credit Opinion on KEDLI, September 2018;
Exhibit___(SFP-10) KEDNY/KEDLI Implied Credit
Metrics;
Exhibit___(SFP-11) S&P Corporate Rating
Methodology, Updated April 1, 2019;
Exhibit___(SFP-12) Moody's Rating Methodology
for Regulated Electric and Gas Utilities, June
23, 2017;
Exhibit___(SFP-13) Debt Cost Calculation;
Exhibit___(SFP-14) Summary of Staff's Cost of
Equity;
Exhibit___(SFP-15) Staff's Proxy Group
Exhibit___(SFP-16) Three-Month Stock Prices for
Staff Proxy Group;
Exhibit___(SFP-17) Discounted Cash Flow
Calculation for Staff Proxy Group;
Exhibit___(SFP-18) Blue Chip Economic
Indicators, March 10,2019;
Exhibit___(SFP-19) Merrill Lynch Quantitative
Profiles;
Exhibit___(SFP-20) Staff Capital Asset Pricing
Model Results;

Exhibit___(SFP-21) Ambika Prasad Dash's Article: "Security Analysis and Portfolio";
Exhibit___(SFP-22) An Excerpt on Sample Size from Dr. Morin's book entitled, Regulatory Finance;
Exhibit___(SFP-23) Illustration of Different Accounting Policy Choices on Operating Income;
Exhibit___(SFP-24) Equity Analysts' Coverage on MGE Energy Inc., El Paso Electric and Otter Tail Corp.;
Exhibit___(SFP-25) Value Line Investment Survey, Quality Control Procedures;
Exhibit___(SFP-26) Energy Information Association (EIA): Annual Energy Outlook, January 2019;
Exhibit___(SFP-27) Generic Finance Case's DCF Results Recast;
Exhibit___(SFP-28) Trends in Utility Stock Valuation and Dividend Yields;
Exhibit___(SFP-29) Wall Street Journal Article entitled, "Can Low Rates Explain High Stock Prices?";
Exhibit___(SFP-30) "The Cost of Equity - A Practitioner's Guide (2010 Edition)" by David C.

Parcell;

Exhibit___(SFP-31) Wall Street Journal Article on Federal Reserve interest rates, dated July 19, 2019;

Exhibit___(SFP-32) Ms. Bulkley's Market Return Using the Federal Energy Regulatory Commission's (FERC) Approach;

Exhibit___(SFP-33) Article by Kent Baker et al. entitled "Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice,";

Exhibit___(SFP-34) "Whose Beta is Best?", Ms. Diana Harrington;

Exhibit___(SFP-35) Impact of Correcting the Flaws in Ms. Bulkley's ROE Approach;

Exhibit___(SFP-36) Regulatory Research Associates' (RRA) Report on State Regulatory Evaluations;

Exhibit___(SFP-37) RRA's, Report on New York State Regulatory Environment;

Exhibit___(SFP-38) Moody's Credit Opinion for National Grid USA, July 2017;

Exhibit___(SFP-39) Commission's Chair's Meeting with RRA dated October 2014;

Rate of Return

Summary

Q. Please summarize your overall rate of return recommendation for KEDNY and how it varies from the Company's request.

A. We are recommending an overall after-tax rate of return of 6.08% for KEDNY, as opposed to KEDNY's request of 6.76%. On a pre-tax basis, our recommended overall rate of return is 7.58% compared to KEDNY's request of 8.53%. The difference between our recommended rate of return and KEDNY's is mainly attributable to a reduction in the cost of equity from 9.65% to 8.20%.

Q. Please summarize your overall rate of return recommendation for KEDLI and how it varies from the Company's request.

A. We are recommending an overall after-tax rate of return of 6.04% for KEDLI, as opposed to the Company's request of 6.74%. On a pre-tax basis, our recommended overall rate of return is 7.54% compared to KEDLI's request of 8.51%. The difference between our recommended rate of return and KEDLI's is attributable to a reduction in the cost of equity from 9.65% to 8.20%.

Q. Please explain the revenue requirement impacts of your adjustments to the Companies' requested rates of return?

A. In terms of revenue requirement, the difference between Staff's and KEDNY's pre-tax rate of return (PTROR) and after-tax rate of return (ATROR) account for approximately $49.0 million and $34.0 million, respectively. Similarly, the difference between Staff's and KEDLI's PTROR and ATROR account for approximately $31.4 million and $22.8 million, respectively. For KEDNY, 10 basis point in ROE is worth nearly $3.3 million and 100 basis points in equity ratio is worth about $5.0 million. For KEDLI, 10 basis point in ROE is worth approximately $2.1 million and 100 basis points in equity ratio is worth about $3.1 million. Our recommended rates of return for both Companies are summarized in Exhibit___(SFP-2).

Fair Rate of Return

Q. Please elaborate as to what is meant by a fair rate of return.

A. A fair rate of return for a regulated utility is

the overall rate of return that enables the company to provide safe and adequate service to its customers, while at the same time assuring the utility continuing support in the capital markets for both its long-term debt and common equity securities at terms that are reasonable given the Company's risk. The overall rate of return is alternatively referred to as a weighted cost of capital.

Q. How is the rate of return typically calculated?

A. The rate of return for a utility company is calculated through a weighted average of the individual cost components of its expected capitalization during the rate year. Typically, there are three sources of capital. The two primary sources are long-term debt and common equity. Customer deposits, while a very small component, is generally reflected in a company's capitalization because it is a relatively permanent and stable source of capital employed by utilities. Investors in debt securities enter into contractual obligations with the utility and receive relatively fixed income streams. Common equity investment is non-contractual. Common

equity investors may share in, but are not guaranteed, a portion of the utility's residual earnings. A fair rate of return allows the utility company to recover its incurred cost of long-term debt while providing its common equity investors the opportunity to earn a return that is comparable to the return available with investments of similar risk.

Q. How are the cost rates of the components of the capital structure typically calculated?

A. The cost rates associated with the company's long-term debt and customer deposits are relatively simple to determine. The costs of existing long-term debt can be readily calculated by examining their contractual terms; i.e., the interest payment on the long-term debt and the amortization of issuance costs. The costs of any new long-term debt instruments, however, require estimation. The cost rate for customer deposits is simply a matter of applying the interest rate that is currently prescribed by the Commission. The cost of common equity, however, is neither contractual nor generically prescribed by the Commission. Its calculation is further

complicated by the fact that all variables affecting a company's cost of equity cannot be directly observed, and instead, some must be estimated.

Q. Is the cost of common equity typically more expensive than the cost of debt for a utility?

A. Yes. Even though both lenders and investors supply the utility with the funds it needs to build and operate its systems, the equity investors only earn a return after the payment of all other expenses. Because equity investors run the risk that a company's achieved returns will not equal their expectations, the return required by equity investors is usually higher than that of the utility's debt holders. Importantly, equity returns are profits and in a company's revenue requirement profits are taxed at the corporate level, which is an added cost to ratepayers that adds to the expense of the equity return.

Q. How can a utility's cost of common equity be measured?

A. The return requirements of a utility's common equity investors can be estimated through a cost of equity analysis. Historically, the Commission

has favored market-based methodologies such as the Discounted Cash Flow (DCF) and the Capital Asset Pricing Model (CAPM) to estimate the return required by equity investors.

Capital Structure

Q. Please discuss the corporate structure of KEDNY and KEDLI.

A. KEDNY and KEDLI are both direct subsidiaries of National Grid USA, which is an indirect subsidiary of the ultimate parent, National Grid Plc. (National Grid), a United Kingdom-based utility holding company. As illustrated in Exhibit___(SFP-3), there are two subsidiary tiers between KEDNY and KEDLI and National Grid.

Q. Do the Companies propose that their rates be set using their stand-alone capitalizations or the capitalizations of either National Grid Plc. or National Grid USA?

A. The Companies propose to use their stand-alone capital structures and propose capital structures that reflect a 48.0% common equity ratio.

Q. Did the Companies provide any rationale for proposing their stand-alone capital structure for

the rate year?

A. Yes. The Companies' capital structure witness, Mr. Jonathan Cohen and cost of equity witness, Ms. Ann E. Bulkley present two main reasons why it is appropriate to use the Companies' stand-alone capital structure for ratemaking purposes. First, the companies believe that they implemented strong ring-fencing measures that have been recognized by the major rating agencies. Secondly, the Companies believe that utility operating subsidiaries of their proxy group have for the last four years maintained average common equity ratios more than 8.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding.

Q. What is your response to the Companies first rationale that they have strong ring-fencing measures that have been recognized by the major rating agencies?

A. We agree with the Companies that they have implemented adequate ring-fencing measures that have been acknowledged by both Moody's and Standard and Poor's.

Q. Could you please elaborate on the ring-fencing measures implemented by the Companies?

A. Yes. As a condition of accepting the Commission's approval of the merger between National Grid and the Companies in 2006, both KEDNY and KEDLI agreed to a series of ring-fencing provisions. These provisions, as well as those agreed to by other New York State utilities, are contained in Exhibit___(SFP-4). We will discuss the specific provisions of the ring-fencing measures later in our testimony.

Q. Please discuss Moody's recent opinion about the strength of the Companies ring-fencing measures.

A. Contained in Exhibit___(SFP-5) is Moody's April 8, 2019, Credit Opinion for KEDNY. The last paragraph of the report states that "[w]hile there is significant debt at KEDNY's parent holding companies, the strong ring-fencing provisions applicable to New York utilities reduce the potential for debt to be pushed down into KEDNY. In particular, we view the explicit leverage restriction for KEDNY (to maintain a debt to capitalization ratio of less than 56%) as providing some credit support at the current

rating level. This provision compares favorably against other New York utilities within the National Grid group. Additional ring-fencing provisions imposed by the NYSPC for KEDNY, which we view as credit supportive, include a special preferred share provision that reduces the probability of bankruptcy in a distressed situation, and the requirement for KEDNY to hold an investment-grade rating." Moody's expressed similar view about KEDLI's ring-fencing measures by stating that "[while] there is considerable debt at KEDLI's parent holding companies, the strong ring-fencing provisions applicable to New York utilities reduce the potential for debt to be pushed down to KEDLI. In particular, we view the explicit leverage restriction for KEDLI (to maintain debt/capitalization at less than 58%) as providing the greatest credit support. Additional ring-fencing provisions imposed by the NYSPC on KEDLI that we view as credit supportive include: (1) the special preferred share provision that reduces the probability of bankruptcy in a distressed situation, and (2) the requirement for KEDLI to hold an investment-grade

rating." This Moody's report published on April 8, 2019, is contained in Exhibit__(SFP-6).

Q. Comparatively, to what extent has Standard and Poor's recognized that the financial protections adequately insulate the Companies from their parent and affiliates?

A. In its latest credit opinions about KEDNY and KEDLI issued separately on September 24, 2018, S&P stated that "…we assess [The Brooklyn Union Gas Co and KeySpan Gas East Corporation] as insulated, with existing insulation measures supporting a one-notch separation between the subsidiary and its parent. Key insulating measures include: a) Prohibition of dividend distributions if the company does not maintain investment-grade credit ratings with at least two of the three major rating agencies, b) Prohibition of dividend increases and payments if the company's debt-to-capital ratio exceeds a certain amount, 3) A golden share whose vote is required to file the utility into a voluntary bankruptcy, 4) Clear economic and strategic incentive from the parent to maintain [The Brooklyn Union Gas Company's] financial

strength."

Q. What is your conclusion regarding the financial independence of KEDNY and KEDLI?

A. As stated earlier, we believe that the rating agencies still view KEDNY and KEDLI as adequately ring-fenced and thus rate the Companies' credit quality substantially on a stand-alone basis, apart from their holding Companies.

Q. Please elaborate on the Companies' second rationale that utility operating subsidiaries with similar business risk characteristics to KEDNY and KEDLI have for the last four years maintained average common equity ratios more than 8.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding.

A. On page 98 of her direct testimony, Ms. Bulkley stated that the average common equity ratio (based on SNL Financial, a business unit of S&P Global, data source) of the utility subsidiaries for the years 2014-2017 was 56.65% for her Combined Utility Group and 57.94% for her Natural Gas Proxy Group.

Q. Do you have any comments about Ms. Bulkley's use of the 56.65% and 57.94% historical common equity ratios to support the reasonableness of the Companies' 48% common equity ratio proposal?

A. Yes. First of all, Ms. Bulkley's use of operating company equity ratios is inappropriate. Investors in her proxy group companies base their return requirements on the holding company's use of leverage, along with its other risk attributes, as they are unable to directly invest in the subsidiaries. Secondly, the capital structures for utility subsidiaries of holding companies may not reflect either rational capitalization policies, or actual common equity employed, and therefore may not be suitable for supporting an operating subsidiary's rate of return. The subsidiary's common equity balance may not, in fact, be financed by common equity at the holding company level. Rather, some of the utility common equity balance may instead be proceeds from debt issued at the holding company level and classified on the utility subsidiary's books as common equity at the time the proceeds were invested in the utility subsidiary.

Q. You stated that the relevant equity ratio to investors is that of the utility holding companies comprising Ms. Bulkley's proxy groups. Have you performed any analysis to ascertain the average common equity ratio of the Companies' witness' proxy group?

A. Yes. According to S&P Capital IQ Financial data, we found that the average common equity ratio of Ms. Bulkley's Combined Utility Proxy Group and Natural Gas Proxy Group for years 2014-2017 were 47.89% and 53.14%, respectively as shown in Exhibit___(SFP-7). This is informative because investors cannot invest in the equity of a wholly-owned subsidiary company. As such, equity investors look at, among other things, the overall capital structure of the consolidated parent company to determine their risk/reward outlook and make investment decisions.

Q. What are the proportions of common equity that the Companies project in the rate year and how were they developed?

A. Per the Companies witness Cohen's Exhibit___ (JC-1, Schedule 2), the rate year proportions of common equity for KEDNY and KEDLI were forecasted

by starting with their respective 2018 year-end capital structures and reflecting the impacts of their forecasted sources and uses of funds from the beginning of 2019 through the rate year ending March 31, 2021. In developing KEDNY's forecasted capital structure with a 51.3% common equity ratio for the rate year, KEDNY indicated that it plans to retain a portion of its earnings and use those earnings to address a portion of its funding needs, driven primarily by capital expenditures of approximately $866 million in the rate year. Further, there is also a $500 million planned equity infusion from the ultimate parent, National Grid during the bridge period. KEDNY plans to issue $1.0 billion of long-term debt during the bridge period and another $400 million long-term debt near the end of the rate year. For its customer deposits balance, KEDNY forecasts an amount of $21.055 million for the rate year compared to $23.413 million as of December 2018.

Q. How did KEDLI develop its forecasted 48.9% common equity ratio for the rate year.

A. In developing KEDLI's forecasted capital

structure with a 48.9% common equity ratio for the rate year, the Company indicated that it plans to maintain a majority of its projected earnings and use them to address its funding needs, which will be driven primarily by capital expenditures of approximately $586 million during the rate year. Unlike KEDNY, KEDLI did not forecast any equity infusion or long-term debt issuance during the bridge period and the rate year. For its customer deposits balance, KEDLI forecasts an amount of $20.581 million for the rate year compared to $14.962 million as of December 2018.

Q. Are the Companies requesting their forecasted common equity ratios be used for ratemaking purposes?

A. No, in order to mitigate their rate requests, the Companies propose to set rates based upon a capitalization with a 48.0% common equity ratio; the common equity ratio upon which their rates are currently set.

Q. What is the Commission's policy on the appropriate capital structure to be used in ratemaking when a utility resides in a holding

company?

Q. The Commission has expressed a preference for using the capital structure of the utility's parent. The Commission elaborated its position in Case 05-E-1222, stating:

"The established regulatory practice in New York in fully litigated rate proceedings, like this one, is to use the consolidated capital structure of the holding parent company for ratemaking purposes. This practice has typically been applied to utility holding company structures in other regulated industries. Holding companies owning electric utilities have only recently appeared in New York and we must now consider whether the conventional approach should apply to the electric industry. We find no valid basis for excluding electric utility holding company structures from the time-honored and well-established regulatory practice. There is no rational basis for us to depart from the approach developed and used for consolidated telecommunication, natural gas and water companies when addressing the substantially similar issues pertaining to consolidated

electric company operations."

Q. Are there circumstances under which the Commission would allow the use of a stand-alone capitalization?

A. Yes. The Commission has indicated that it will consider using a stand-alone capitalization if a utility can demonstrate that it has adequately isolated itself from the risks of its affiliates.

Q. How can a utility demonstrate that it has isolated itself from the risk of its affiliates?

A. In its Abbreviated Order Authorizing Acquisition, issued September 9, 2008, and its Order Authorizing Acquisition Subject to Conditions, issued January 6, 2009 in Case 07-M-0906 where the Commission approved Iberdrola's acquisition of Energy East, the Commission stated it would consider using a stand-alone equity ratio upon a showing that the rating agencies consider the utility operating subsidiary in question adequately insulated from the risks of the parent company's other operations. A company can establish such insulation by demonstrating the presence of significant ring-fencing measures.

Q. Can you briefly explain the concept of ring-

fencing in this context?

A. Yes. Ring-fencing, with respect to a regulated public utility subsidiary, occurs when a subsidiary financially separates itself from its holding company/parent company to protect the subsidiary from financial instability or the bankruptcy of the parent. The intent of the ring-fencing measures is to insulate assets in the subsidiary from the risks of the holding company/parent and from the other subsidiaries in the holding company. In theory, if a subsidiary is ring-fenced, and the holding company or another subsidiary goes bankrupt, creditors would not be able to attach their claims to the assets of the ring-fenced subsidiary. In addition, a fully ring-fenced company should be insulated from its parent's influence regarding the financial decisions of the subsidiary. In theory, ring-fencing protects the assets of the subsidiary in the event of a bankruptcy of a parent or other affiliate. Moreover, ring-fencing may allow a subsidiary to be rated higher than the parent.

Q. Besides bankruptcy protection, what are the

advantages of ring-fencing a utility subsidiary?

A. If a utility subsidiary achieves a higher rating than its parent, there is a benefit of a lower cost of debt to the subsidiary and ultimately to ratepayers.

Q. Does the existence of adequate ring-fencing require the use of the subsidiaries' stand-alone capital structures in establishing the ratemaking capital structure?

A. No, it does not. While the presence of adequate ring-fencing provides the opportunity to use the utility's stand-alone capital structure, the financial policies of the utility must still be evaluated with an eye toward identifying whether a suitable financing mix, resulting from reasonable financing policies, is being deployed. The idea is to find the approximate ratios of debt and equity that lead to the lowest weighted cost of capital.

Q. Please describe what constitutes reasonable financial policies for a regulated utility.

A. A reasonable financial policy is one that provides an optimal mix of debt and equity financing. An optimal financing policy balances

the interests of both ratepayers and investors by having the lowest cost of capital for an acceptable level of risk. A regulated utility, through its financial policies, should obtain a balanced mix of capital sources that will result in the most economical financing of its assets over the long run. For a utility company, a balanced capital structure will contain a sufficient proportion of common equity that will enable the utility to raise capital most economically. While common equity is costlier than debt, it should still be employed in sufficient amounts that will provide an economically efficient credit rating.

Q. How is an economically efficient credit rating determined?

A. An economically efficient credit rating is one that provides the lowest long-run cost of capital. The Recommended Decision in Case 91-M-0509 regarding Generic Financing generally determined that the overall cost to ratepayers, as measured by the pre-tax rate of return, was minimized at either a "BBB" or "A" bond rating. When practical, the Commission has generally

supported rate levels that provide utilities the opportunity to achieve and maintain a bond rating within the "A" range.

Q. What credit ratings are assigned to KEDNY and KEDLI by Moody's and S&P?

A. As indicated in our previously referenced Exhibit___(SFP-5) and Exhibit___(SFP-6) as well as Exhibit___(SFP-8) and Exhibit___(SFP-9), Moody's assigns a credit rating of "A3" to KEDNY and KEDLI. S&P assigns a rating of "A-" to both Companies.

Q. Have their "A3/A-" credit ratings provided the Companies ready access to the capital markets at reasonable rates?

A. Yes. KEDNY's access to the capital market is evidenced by its February 2019 issuance of $550 million of 10-year fixed rate senior unsecured securities and $450 million of 30-year fixed rate senior unsecured securities at very favorable rates of 3.865% and 4.487% respectively. KEDLI has not issued any long-term debt securities since 2016. As previously discussed, KEDLI has the same "A3/A-" rating as KEDNY and is viewed as possessing very similar credit risk attributes,

and thus, should have similarly strong access to the credit markets.

Q. What capital structure do you propose for the Companies?

A. We accept the same 48% common equity ratio proposed by the Companies.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDNY?

A. We based the capital components on Staff's determination of Staff's adjusted earnings base for KEDNY. To Staff's adjusted earnings base of $4,834,497,000, we applied the 48% common equity ratio to determine a common equity component of $ 2,320,559,000 ($4,834,497,000*48%) in our recommended capital structure. We included in our recommended capital structure KEDNY's projected customer deposits balance of $21,055,000 for the rate year. After determining the common equity and customer deposits components, the residual of $2,501,430,000, or approximately 51.56% of Staff's adjusted earnings base, represents the projected long-term debt component for KEDNY.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDLI?

A. We based the capital components on Staff's determination of Staff's adjusted earnings base for KEDLI. To Staff's adjusted earnings base of $3,269,060,000 we applied the 48% common equity ratio to determine a common equity component of $1,569,149,000 ($3,269,060,000*48%) in our recommended capital structure. We included in our recommended capital structure KEDLI's projected customer deposits balance of $20,581,000 for the rate year. After determining the common equity and customer deposits components, the residual of $1,679,330,000, or approximately 51.37% of Staff's adjusted earnings base, represents the projected long-term debt component for KEDLI.

Q. Do you believe that your capital structure recommendation will be supportive of the Companies' financial integrity?

A. Yes, we believe that our capital structure recommendation will be supportive of the Companies' financial integrity.

Q. Have you performed any analyses to substantiate your assertion that your recommended capital structure with a 48% common equity ratio, together with your recommended return on common equity (ROE) will be supportive of the Companies' financial integrity?

A. Yes, our recommended common equity ratio and return on equity recommendation when combined with other quantitative metrics and qualitative factors produce credit metrics that are generally consistent with the performance required by Moody's and S&P to maintain the Companies' current ratings of "A3" and "A-" respectively, as shown in Exhibit___(SFP-10).

Q. Please discuss how you determined that the Companies' credit metrics are consistent with those ratings.

A. We based our analysis on the overall credit metrics used by Moody's and S&P in evaluating both the financial and business risks of regulated electric and gas utilities. We note that only S&P provides specific categories of financial and business risks.

Q. Please discuss the concept of financial risk.

A. Financial risk is the additional risk a shareholder bears when a company uses debt in addition to equity financing. Companies that employ greater leverage would have higher financial risk than companies financed mostly or entirely by equity. S&P financial risk profile categories include “Minimal”, “Modest”, “Intermediate”, “Significant”, “Aggressive”, and “Highly Leveraged” based on seven different financial metric ratios. Reflected in Exhibit___(SFP-11) is the S&P Corporate Rating Methodology report originally published on November 19, 2013, and most recently updated on April 1, 2019. As illustrated on pages 33 and 34 of the report, S&P calculates two core coverage ratios and six supplementary coverage ratios. Specifically, S&P examines the following financial risk ratios in assessing a firm’s credit risk: a. Funds from Operations(FFO)/debt (core metric); b. Debt/EBITDA (core metric,); c. Cash Flow from Operations (CFO)/debt; d. Free Operating Cash Flow (FOCF)/debt; e. Discretionary Cash Flow (DCF)/debt; f. (FFO + interest)/cash interest (FFO cash interest cover); and g.

EBITDA/interest.

Moreover, based on the methodology guidelines for the above metrics, we used S&P's medial volatility financial benchmark table reflecting the Companies lower-risk, rate-regulated distribution operations.

Q. In addition to the financial risk metrics, did you consider other qualitative factors considered by S&P in its business risk analysis of the Companies?

A. Yes, we did. S&P's business risk profile analysis incorporates such factors as country risk, industry risk, regulatory environment, company competitive position, business and geographic diversification, and management strategy. S&P's business risk profile categories include "Excellent", "Strong", "Satisfactory," "Fair", "Weak", and "Vulnerable". In determining the "A-" rating, S&P has assigned a business risk profile of "Excellent" to the Companies. Therefore, we used an "Excellent" business risk profile in analyzing the credit metrics of the Companies.

Q. Please discuss S&P's rationale for assigning an

"Excellent" business risk profile to KEDNY?

A. As illustrated in previously referenced Exhibit___(SFP-8), S&P credits the low operating risk of KEDNY's regulated gas distribution operations to its predominantly residential and commercial customer base, economically robust service territory and generally constructive regulatory environment as the primary drivers of the "Excellent" business risk evaluation.

Q. What is S&P's rationale for assigning an "Excellent" business risk profile to KEDLI?

A. As shown in our Exhibit___(SFP-9) referenced earlier, S&P's "Excellent" business risk assessment for KEDLI is based upon the Company's low operating risk regulated distribution operations, its somewhat small service territory that lacks geographic and operating diversity, a predominately residential and commercial customer base, and a generally constructive regulatory framework.

Q. Please continue with your credit metrics analysis based on Moody's rating guidelines.

A. Our Exhibit____(SFP-12) contains the June 23, 2017, Moody's ratings methodology for regulated

electric and gas utilities. Page 4 of the Moody's report contains the following financial strength metrics examined by Moody's: a) Cash Flow from Operation (CFO), pre-Working Capital (WC) + Interest/Interest, with Moody's assigned weight of 7.5%; b) CFO pre-WC/Debt, with Moody's assigned weight of 15%; c) CFO pre-WC-Dividends/Debt, with Moody's assigned weight of 10%; and d)Debt/Total Capital, with Moody's assigned weight of 7.5%.

Q. Did you consider the Companies' business profile/qualitative factors (business risk profile) based on Moody's rating guidelines?

A. Yes, we did consider the business risk profile of the Companies based on Moody's rating guidelines for regulated electric and gas utilities as shown in our Exhibit___(SFP-12) referenced earlier. Moody's business risk profile analysis, which accounts for 60% of their overall indicative grid rating incorporates such factors as: 1) Legislative and Judicial Underpinnings of the Regulatory Framework; 2) Timeliness of Recovery of Operating and Capital Costs; 3) Consistency and Predictability of Regulation; 4) Sufficiency

of Rates and Returns; 5) Market Position; and 6) Generation and Fuel Diversity. Each of the six business risk factors is ranked as "Aaa", "Aa", "A", "Baa", "Ba", "B", and "Caa". Given that the Companies are rated by Moody's, we used Moody's qualitative scores for the Companies averaging close to "A3".

Q. How did you apply the 11 credit metrics consisting of Moody's four financial metrics and seven S&P financial benchmarks?

A. In our previously referenced Exhibit___(SFP-10), we applied the 11 credit metrics using Companies financial results based upon our capital structure, cost of debt, and cost of common equity recommendations. As the exhibit illustrates, the overall financial metrics when combined with the "Excellent" business risk profile of the Companies, produce credit metrics consistent with the rating guidelines of S&P rating of close to "A-" and Moody's rating of "A3".

Q. For comparison purposes, did Mr. Cohen provide a projection of the Companies' financial metrics based on the Companies' requested capital

structure and cost rates?

A. Yes, the Companies' response to IR DPS-590 provided the rate year credit metric projections. According to the Company's projections, KEDNY's forecasted core and supplementary credit metrics are generally similar to what KEDNY has historically maintained over the past three years. With the exception of CFO pre-WC-Dividends/Debt, KEDLI's other forecasted metrics will result in higher core and supplementary metrics than it has historically exhibited. As explained earlier, these credit metrics are based on the medial volatility/low-risk grid table of S&P/Moody's reflecting the low business risk profile of the Companies.

Q. Do you have any observations about the Companies response to IR DPS-590?

A. Yes. We noted that some of the inputs for the credit metrics contained in the IR DPS-590 are not consistent with the Companies filing. For instance, the common equity ratios in IR DPS-590 for KEDNY and KEDLI are approximately 63% and 62%, respectively compared to the 48% in their rate year request. Also, the Net Pension & Other

Postemployment benefits (OPEBs) accrual & deferral for KEDNY and KEDLI are negative $1.64 million and negative $4.76 million respectively, compared to positive $47.47 million and negative $36.39 million in the rate filing. Due to such inconsistencies, we recalculated the Companies credit metrics based on their filing and rating agencies adjustments provided by the Companies in their response to IR DPS-590 as shown in our previously referenced Exhibit___(SFP-10).

Q. How do your financial metrics and implied credit ratings compare with the recalculated financial metrics and ratings in the Companies' filing?

A. For KEDNY, our previously referenced Exhibit___(SFP-10) shows coverage ratios and implied "A2"/"close to A-" Moody's/Standard & Poor's credit ratings compared to "A2"/"A-" implied by the Company's filing. Similarly, for KEDLI, our implied Moody's/Standard & Poor's credit rating is "A3"/"close to A-" compared to "A3"/"close to A-" in the Company's filing. One core coverage ratio used by S&P to assess the relative financial risk of regulated electric and gas utility companies is leverage ratio (Debt to

EBITDA—Earnings Before Interest, Taxes, Depreciation and Amortization). The ratio indicates a company's ability to cover its fixed debt service (interest plus principal) obligations as well as the variability of its earnings net of interest payments. The lower the leverage ratio the better. As the exhibit illustrates, our fallout Debt to EBITDA coverage ratio is 5.13x and 4.99x for KEDNY and KEDLI, respectively compared with the implied 4.86x and 4.53x for KEDNY and KEDLI, respectively, as calculated from the Companies' filings. Per S&P's rating guidelines for regulated utilities, the appropriate benchmark for a regulated electric and gas utility in the single "A" range with an excellent business risk profile is between 2.5x-4.5x. Another core metric used by S&P to assess the financial health of regulated electric and gas utilities is Funds from Operation to Debt (FFO/debt). The higher the FFO/Debt ratio the better. Under this metric, our capital structure recommendation along with Staff's other adjustments produces FFO/Debt ratio of 19.4% and 14.1% for KEDNY and KEDLI,

respectively compared with 21.9% and 16.4% for KEDNY and KEDLI, respectively in the Companies' filings. The acceptable FFO/Debt benchmarks reasonably expected from a regulated electric and gas utility in the single "A" range with an excellent business risk profile is between 13-35% per S&P rating guidelines.

Q. Please continue with your core metric analysis relating to Moody's guidelines.

A. The core coverage ratio used by Moody's to assess the financial health of regulated electric and gas utility companies is Cash flow from Operation Pre-Working Capital to debt (CFO pre-WC/Debt). The ratio indicates a company's ability to cover its fixed debt service (interest plus principal) obligations as well as the variability of its earnings net of interest payments. The higher the coverage metric the better. As our previously referenced Exhibit___(SFP-10) illustrates, our fallout (CFO pre-WC/Debt) metric is 19.2% and 13.6% for KEDNY and KEDLI, respectively compared with the implied 21.7% and 15.9% for KEDNY and KEDLI, respectively, as calculated from the Companies' filings. Per

Moody's rating guidelines for regulated utilities, the appropriate benchmark for a regulated electric and gas utility in the single "A" range with a low business risk profile is between 19%-27%. While KEDLI's underperformance on this metric is of potential concern, we note that Moody's will often discount performance in the short run particularly if it is related to cash flow impacts that may not be durable. For instance, during fiscal years 2016 through 2018, KEDNY's CFO pre-WC/Debt was below the 13.7% figure forecast for KEDLI during the rate year. Due to higher than anticipated environmental remediation costs KEDNY only achieved successive CFO pre-WC/Debt ratios of 7.9%, 13.3% and 9.8% during this timeframe and was able to maintain an "A2" rating. In the unlikely event that Moody's downgrades KEDLI, there would be no immediate rate impact because the Company has no plans to issue new debt until after the end of the rate year.

Q. Did you review the most recent Moody's and S&P credit analyses to ascertain each rating agency's view with respect to the Companies' ratings

outlook and the factors each indicate could lead to a rating upgrade or downgrade?

A. Yes. As illustrated in our previously referenced Exhibit___(SFP-10) and Exhibit___(SFP-11), in September 2018, S&P's assessment indicated a "Stable" outlook for KEDNY's and KEDLI's "A-" rating. According to S&P, a rating outlook assesses the potential direction of a long-term credit rating over the intermediate-term (typically six months to two years). Therefore, it does not view KEDNY's and KEDLI's "A-" credit rating as likely to change over the intermediate-term.

Q. What is Moody's recent assessment of the Companies' outlook?

A. In its April 2019 credit opinion report, as shown in our previously referenced Exhibit___(SFP-5) and Exhibit___(SFP-6), Moody's assessment indicated a "Stable" outlook for the Companies' "A3" rating. Therefore, Moody's does not view KEDNY's and KEDLI's "A3" credit rating as likely to change over the intermediate-term. Moody's indicated that the Companies could be downgraded if their respective CFO pre-WC/debt is

consistently below the high teens.

Q. The Companies' capital structure witness, Mr. Cohen, indicated that there has been pressure on both the Companies credit metrics and bond ratings in recent years thereby putting KEDNY/KEDLI's objective of maintaining an "A3" bond/credit rating at risk. As a result, in February 2019, Moody's downgraded the Companies ratings by one notch to "A3" from "A2". Do you have any comment?

A. Yes. First, Moody's downgraded both Companies rating from an "A2" to "A3" in February 2019 largely because of the impact of the Federal Tax Cuts and Jobs Act (Tax Act) of 2017, as well as heightened capital investments, and in the case of KEDNY also because of higher spending on environmental remediation. The recent downgrade of the Companies by Moody's was not unique, the reduced cash flows resulting from the Tax Act led Moody's to downgrade many utilities over the past year and a half. As we have demonstrated, the Companies could maintain their current "A3" credit ratings with our implied credit metrics and access the capital markets at reasonable

rates. Moreover, both rating agencies (Moody's and S&P) have placed the Companies' current ratings on a "Stable" outlook. Per these rating agencies, a rating outlook assesses the potential direction of a long-term credit rating over the intermediate-term (typically six months to two years). Therefore, KEDNY's and KEDLI's "Stable" outlook suggests that the Companies' rating is not likely to change over the intermediate-term.

Cost of Long-Term Debt

Q. What costs of debt are the Companies requesting in this proceeding?

A. In its filing, KEDNY requested a rate year cost of debt of 4.11%. We came across an error in the calculation of KEDNY's long-term debt cost rate which was acknowledged by the Company in response to an IR number DPS-1002 dated August 21, 2019. Therefore, the Company's corrected debt cost is 4.14%. KEDNY's 4.14% cost rate reflects the weighted average cost of its outstanding long-term debt as well as a projected 3.83% interest rate for a planned $400 million issuance of 10-year senior unsecured debt obligations near the

end of the rate year. KEDLI is requesting a rate year cost of debt of 4.06% reflecting its embedded debt cost. KEDLI has no plans to issue any new long-term debt during the link period and the rate year.

Q. How did KEDNY determine the 3.83% interest rate for the $400 million planned debt issuances during the rate year?

A. Per Mr. Cohen's Exhibit___(JC-1), page 2 of Schedule 1, KEDNY determined the 3.83% interest rate for the new debt security by adding a 117 basis point spread associated with KEDNY's recent debt issuance in February 2019 to a 2.66% forward rate of US 10-year Treasury for March 1, 2021.

Q. Are you proposing any adjustments to the Companies' requested cost rates?

A. Yes. We are proposing one adjustment to KEDNY's request and no adjustments to KEDLI's calculation.

Q. Please discuss your adjustment to KEDNY's requested rate year cost of debt.

A. Rather than using the 10-year Treasury forward rate recommended by Mr. Cohen, we recommend using the most recent (June 2019) 10-year Treasury

yield of 2.07% in combination with the Company's 117 basis points spread to estimate an interest rate of 3.24% for KEDNY's proposed new debt issuance as shown in our Exhibit___(SFP-13). The interest rates should be updated based upon interest rates and yield spreads in existence just prior to the Commission's decision.

Q. Why did you use the most recent 10-year Treasury yield to estimate the interest rate for KEDNY's new debt issuance?

A. We used the most recent 10-year Treasury yield in this rate proceeding because the Commission considers current interest rates to be the best predictor of where interest rates will be in a given rate year going forward. Specifically, in Case 10-E-0050, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Electric Service, the Commission stated that, "[i]n as much as we are making a one-year rate determination, the current interest rates are reasonably representative for the immediate future."

Q. Did you make changes to KEDNY's proposed 4.14%

long-term cost rate for the rate year?

A. No. Although we used a 3.24% interest rate for KEDNY's projected new debt issuance, the fact that those funds will be raised so late in the rate year, there was only a di minimis impact on the overall debt cost rate of 4.14% proposed by KEDNY. As illustrated in Exhibit___(SFP-13) referenced earlier, we are accepting the 4.14% cost of debt for the rate year ending March 31, 2021.

Cost of Customer Deposits

Q. What customer deposits rate are the Companies proposing for the rate year ending March 31, 2021?

A. The Companies are proposing a 2.45% customer deposits rate currently established by the Commission for investor-owned utilities in Case 18-M-0611.

Q. Do you agree with the Companies' customer deposits rate proposal?

A. Yes, we agree, however, prior to the Commission decision in this case we also recommend that the customer deposits rate be updated to reflect any

new rate established by the Commission.

Cost of Common Equity

Summary

Q. Please summarize your analysis in determining your 8.20% cost of common equity.

A. Our common equity cost estimate is based on applying a DCF analysis and CAPM analysis to a proxy group of 28 electric utility holding companies. The DCF applied to the proxy group results in an average equity cost of 8.22%. We averaged two different CAPM analyses to produce an equity cost of 8.04%. Following the Commission's practice, we apply weightings of two-thirds to the DCF and one-third to the CAPM, which resulted in a ROE of 8.16% which we rounded up to 8.20%. A summary of these calculations can be seen in Exhibit___(SFP-14).

Q. Please discuss prior Commission precedent in estimating the cost of equity.

A. The Commission has consistently used the methodology of weighting the DCF result as two-thirds of the total equity cost and the CAPM result as one-third in estimating a utility's

cost of equity. For example, in Case 10-E-0362 involving Orange & Rockland Utilities, Inc. and in Case 10-E-0050 involving Niagara Mohawk electric rate case, the Commission authorized a cost of equity based on two-thirds DCF and one-third CAPM methodology. This methodology was also used in Case 16-G-0257, a rate proceeding for National Fuel Gas (NFG). Consistent with Staff's approach and prior Commission precedent, we applied the DCF and the CAPM approach to a proxy group of companies that were selected based on the criteria that we will discuss later.

Staff Proxy Group

Q. Why do you use a proxy group in your analyses to estimate the Companies' cost of equity?

A. The use of a proxy group to determine KEDNY/KEDLI's cost of equity is necessary because the Companies' common stock is not publicly traded, and thus, direct DCF and CAPM analyses of the Companies are not possible. Equally important is that the DCF analyses for individual companies are reliant on equity analysts' estimates of growth which are, by nature,

inaccurate and may also be biased. Similarly, beta determinations used in the CAPM methodology are based on historical observations, that due to circumstances such as corporate restructurings or industry transformations, may not represent the level of earnings volatility expected in the future. By employing a sufficiently large proxy group of similarly situated companies, the undesirable effects of biased (both upward and downward), inaccurate growth estimates or beta measures for any individual company is diminished. While the objective is to choose a group of companies whose risks closely match those of the company being examined, it is important that the group selected is also large enough to provide confidence in its results. The greater the number of suitable companies, the less sensitive the overall cost of equity estimate will be to the effect of any potential inaccuracies in the data for any particular company.

Q. Please describe how you developed your proxy group.

A. Even though the Companies are local gas

distribution companies, their ultimate parent is largely considered to be an electric utility by investors. Accordingly, we believe that the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be "electric utilities." The majority of these companies have operating subsidiaries that are combination electric and gas utilities and thus our analysis with this group will reflect the challenges and opportunities facing energy utilities in general. We believe that the two types of utilities are fundamentally very similar. Both are regulated public utilities that are subject to nearly identical ratemaking treatment. In addition, both invest primarily in capital-intensive physical networks that connect the customer to the source of supply; and sell their products and services at regulated rates to customers.

Q. How does your approach compare with the approach Staff uses to develop a proxy group for the Companies sister company Niagara Mohawk Power Corporation (Niagara Mohawk)?

A. Consistent with our approach in this proceeding,

and as Staff has done in all Niagara Mohawk rate proceedings over the past twenty years, Staff in Cases 17-E-0238 and 17-G-0239, employed a proxy group using companies screened from Value Line's electric utility companies.

Q. What criteria did you apply to the Value Line electric utility group?

A. To be selected for our proxy group, a company had to meet the following requirements: (1) an investment grade rating from both Moody's and S&P ("Baa3" and above by Moody's and "BBB-" and above by S&P); (2) regulated revenues that constituted at least 70% of that company's total revenue, as determined by each company's 2018 SEC annual report (10-K); (3) currently paying common stock dividends; and (4) not currently involved in any merger/acquisition (M&A) activity.

Q. Please describe how you selected your proxy group.

A. As shown in Exhibit___(SFP-15), our proxy group consists of 28 companies. Beginning with the 37 holding companies deemed by Value Line to be "electric utilities", we eliminated Avangrid Corporation because it has only been publicly-

traded since December 2015, which we believe is an insufficient period or track record to establish confidence in the financial forecasts needed for calculating a company's cost of equity. Specifically, Value Line usually calculates its value of beta over a five-year period, but Avangrid Corporation has only three years of stock price data from which to calculate beta. Similarly, we also eliminated Evergy, Inc. because it has only been publicly-traded for about a year, a time frame so short that Value Line hasn't attempted to calculate a beta. As for the remaining 35 companies, we eliminated five companies whose regulated revenues constitute less than 70% of total revenue. These companies are CenterPoint Energy Inc., DTE Energy Co., Exelon Corp., Otter Tail Corp and Public Service Enterprise Group Inc. Regarding the remaining 30 companies, we eliminated Fortis Inc. because it is a Canadian-based company, which, although traded on both the Toronto and New York stock exchanges, has its financial reports and Value Line projections based on Canadian currency which would introduce the possibility that

exchange rate variations could affect the DCF calculation. Finally, we eliminated El Paso Electric because of a pending merger/acquisition transaction. Therefore, we arrived at a proxy group comprised of 28 electric utility holding companies which met all four criteria stated previously.

Discounted Cash Flow Methodology

Q. Please describe the DCF methodology.

A. The DCF methodology assumes that the principles used to measure the cost of common equity are the same as those used in measuring the yield investors require on debt. However, while interest payments are known with relative certainty, future dividend payments are uncertain. The foundation of the DCF is that investors will price common stock to equal the present value of future dividend payments. Therefore, the valuation of a share of stock can be represented by the following dividend growth rate expression:

$$P_0 = \frac{D1}{(1+k)^1} + \frac{D2}{(1+k)^2} + \frac{D3}{(1+k)^3} \ldots \frac{Dn}{(1+k)^n}$$

Where:

P_0 = current stock price

D_n = dividend in period "n"

k = cost of equity.

Q. Please describe your DCF methodology and its results.

A. As illustrated in Exhibit___(SFP-16), the DCF uses a three-month average stock price, calculated by averaging the high and low monthly prices for each company in the proxy group. Our analysis covers the period from April 2019 to June 2019. As shown in Exhibit___(SFP-17), our DCF methodology develops a stream of dividends for each proxy company. Each dividend stream is discounted at a rate which makes the discounted value of its dividend stream equal to its respective three-month average stock price. The discount rate required to discount the stream of expected dividend payments into the current stock price is the estimated cost of equity for that company. Our DCF methodology calculates an average return on equity of 8.22% for the proxy

group.

Q. How did you develop your dividend projections?

A. We employed a two-stage DCF methodology. For the first stage encompassing the 2019 through 2023 period, we used Value Line dividends per share estimates. Beyond 2023, we calculated a sustainable growth rate for each company in the proxy group based primarily upon its expected return on equity and projected retention ratio. Also included in the sustainable growth rate estimates is the external growth that can be realized when a company issues new shares at prices above book value.

Q. How did you calculate the short-term dividend growth rates?

A. We calculated the short-term dividend growth rate by solving for the compound annual growth rate between the base year value of our dividend stream of 2019 and the ending year value of our dividend stream in 2023. The median and average short-term growth rates for our proxy group are 5.51% and 5.06%, respectively.

Q. How did you derive the long-term sustainable dividend growth rates used in the DCF?

A. Sustainable dividend growth rates depend on a firm's expected internal and external growth rates. Long-term internal growth will be achieved based upon a company's earned return on common equity and on the portion of earnings retained within the business. That is, internal growth, also known as retention growth, is achieved based upon the growth in a company's retained earnings. In formulaic form, retention growth is measured by the following formula:

Retention growth = b*r

Where:

"b" = the fraction of earnings retained in the company

"r" = the expected rate of return on common equity.

For example, if a firm retains 40% of its earnings and earns 10% on equity, then its retention growth rate will be 4.0% (40%*10%). Thus, the growth rate is determined by the increase in the equity retained in the business and the expected return on that equity. Our analysis calculates median and average retention growth rates of 3.89% and 4.09%, respectively for

the proxy group.

Q. How is external growth measured in your DCF methodology?

A. As discussed earlier, external growth can be achieved if a company issues common equity above its book value. This is commonly referred to as the SV factor, where "S" is the increase in common shares outstanding that have been sold and "V" is the per share premium or discount to the book value on the shares sold. Typically, Staff evaluates the SV factor using the formula: (MBR-1) * growth in common shares, where: (MBR) is the market-to-book ratio. Thus, the formula for the sustainable growth rate is given by g = (b*r) + (SV), where SV is also given by (MBR-1)*growth in common stock. In our analysis, the median and average external growth rates for the proxy group are 0.57% and 0.77%, respectively.

Q. What is your sustainable dividend growth rate for the proxy group?

A. The aggregation of our retention growth and external growth produces median and average sustainable growth rates of 4.64% and 4.86%, respectively.

Q. How did you check the reasonableness of your sustainable growth rate?

A. We compared it with the projected overall growth rate of the economy. Rational utility investors expect growth to generally track the changes in output, or growth in the overall economy, as measured by growth in the nominal Gross Domestic Product (GDP). As reported in the March 10, 2019 edition of Blue Chip Economic Indicators shown in Exhibit___(SFP-18), the consensus long-range estimate of nominal GDP growth, going out to 2026-2030, is 4.0%, which is modestly lower than our median sustainable growth rate of 4.64%. Thus, if anything, our sustainable growth rate is more optimistic on utility growth prospects. It should be noted that the 4.0% nominal GDP growth rate estimate is comprised of two components: real GDP growth of 1.9% and an inflation rate of 2.1%. The long-run projections generally show annual real GDP modestly increasing from a rate of 1.7% in 2021 to a rate of 2.0% in 2025, while inflation is forecasted to hold between 2.1%-2.2% from 2021 and beyond into the long-run. This comparison is appropriate because the

nominal GDP rate reflects assumptions about future inflation, in addition to the real growth expected in the economy as a result of productivity gains. Therefore, it would not be unreasonable for investors in the market, to expect their future dividends to generally keep pace with overall inflation, as well as reflect productivity gains similar to those expected for the economy as a whole. Likewise, for investors in a mature sector of the economy, such as the utility industry with perhaps slower-than-average growth prospects, it is not unreasonable to expect future dividend growth to be roughly equivalent to that of the overall economy.

Q. What was the result of your DCF analysis?

A. Based on the inputs that we have just described, the DCF average cost of equity for our proxy group is 8.22%, while the median result is 8.11%.

Capital Asset Pricing Model

Q. Please explain the concept underlying the Capital Asset Pricing Model.

A. The basic concept underlying the CAPM is that risk-averse investors demand higher returns for

assuming additional risk. In other words, higher-risk securities are priced to yield higher returns than lower-risk securities.

Q. How does the CAPM measure risk?

A. The CAPM quantifies the additional return, or risk premium, required for bearing incremental risk above the risk-free rate of return. The model calculates a cost of equity based upon the premise that an investor who diversifies his security portfolio is only exposed to systematic risk (i.e. market risk). The systematic risk represents the risk of investment within a well-diversified portfolio. The remaining risks are not assumed by a rational investor since these risks can be eliminated through diversification. Hence, investors do not require compensation for diversifiable risks. In the CAPM, the beta of a stock measures the volatility of that stock in relation to the volatility of the stock market as a whole. It is produced by a regression analysis and represents the covariance of a stock with the market. A beta above 1 means the stock has greater risk than the market; a beta below 1 means that a stock has less risk than the market.

A beta of 1 means the risk of the stock is equivalent to the risk of the market.

Q. What inputs are used in your CAPM models and how did you derive them?

A. There are three inputs in the CAPM equation: the risk-free rate, the beta of a stock, and the market risk premium. We employed the monthly average of 10-year and 30-year Treasury bond yields over a three-month period from April 2019 to June 2019, as the risk-free rate. The resulting risk-free rate is 2.55%.

The beta that we employed is the average of the latest Value Line published betas for the 28 utilities in our proxy group. As shown on page 1 of the previously referenced Exhibit___(SFP-17), we have calculated the average Value Line beta for our proxy group to be 0.61.

Q. Why did you use the mean beta rather than the median beta of the proxy group?

A. As a practical matter, there is only a minor difference, as the median beta of the proxy group is 0.60. Employing the median beta rather than the mean beta in this case would not materially change our overall ROE calculation of 8.16%. Our

model produces an 8.16%, however, if we used the 0.60 median beta, the overall ROE falls by about two basis points to 8.14%. Nonetheless, we note that we have previously recommended the use of the median beta because it diminishes the undue influence of any outlying value of beta. In addition, Value Line reports its beta determinations in increments of 0.05, therefore, to be consistent, changes from month to month in the median value should only change in increments of 0.05 as well. Changes in the CAPM results based on the median beta are not as smooth as those based on the mean, because this attribute has the potential to introduce unintended volatility in the determination of the proxy group's beta, therefore, we have employed the mean beta as the appropriate input to the CAPM.

Q. How did you arrive at your 8.33% estimate of the market risk premium?

A. Our estimate of an 8.33% market premium was based on the market return estimates contained in Merrill Lynch's Quantitative Profiles reports less the 2.55% risk-free rate discussed earlier. Specifically, we used the May, June and July 2019

editions of Merrill Lynch's Quantitative Profiles and averaged the required and implied market returns of each of the three point-in-time estimates to arrive at an average required return for the market of 10.88%. Then we subtracted the 2.55% average risk-free rate to arrive at the 8.33% market premium. The Merrill Lynch Quantitative Profiles are attached as Exhibit___(SFP-19).

Q. Why did you use Merrill Lynch data for calculating market risk premiums?

A. Our market risk premium is calculated by subtracting the risk-free rate from Merrill Lynch's cost of equity for the market. The Merrill Lynch data provides two forward-looking returns on the market: a required return and an implied return. Merrill Lynch is a respected independent firm with many resources available to make the calculations. This market risk premium estimation methodology has been used by the Commission since 1996, in Case 95-G-1034, Central Hudson Gas & Electric Corporation - Rates, Order and Opinion No. 96-28 issued October 3, 1996, on page 14, where the Commission approved the use of

the Merrill Lynch estimate. In its Opinion, the Commission stated, "…the Judge's market return calculation based on Merrill Lynch data is a reasonable method of deriving a risk premium."

Q Why did you employ 10-year and 30-year Treasury bond yields as the risk-free rate in your analysis?

A. While there are not any true risk-free rate securities, Treasury securities backed by the full faith and credit of the United States government are very low-risk investments. As such, they serve as an excellent surrogate for the risk-free rate. Additionally, the blending of the yields on Treasury securities with a 10-year and 30-year maturity is reasonable because it approximates most investors' time horizon.

Q. Why did you use Value Line betas in your analysis?

A. Value Line is a well-known publication and its beta calculations are widely available to investors. In addition, the Commission historically uses the Value Line betas in estimating the CAPM cost estimates because Value Line calculates its betas over a five-year

period, thereby mitigating the inherent volatility of using beta estimates calculated over shorter time periods.

Q. Please describe the calculation of the traditional CAPM.

A. The traditional CAPM expression takes the form:

$K = Rf + \beta*(Rm-Rf)$

Where:

K = investor's required return or equity cost of capital

Rf = risk-free rate of return

β = beta, measure of both business and financial risk (systematic risk)

Rm = market rate of return

Rm–Rf = market risk premium

The traditional CAPM determines the return on common equity by adding a company's non-diversifiable risk premium to the risk-free rate. The non-diversifiable risk premium is measured by multiplying the proxy group's beta by the market risk premium.

Q. What is the result of your traditional CAPM analysis?

A. As calculated below, our traditional CAPM cost of

equity is 7.63%:
Required return = (10.88%-(2.55%)*0.61+2.55%

Q. Please describe the zero-beta CAPM.

A. The zero-beta CAPM is a two-factor version of the standard CAPM. The zero-beta CAPM determines the return on equity by adding two factors to the risk-free rate: first, a factor of 0.75%, multiplied by the average beta of the proxy group and the S&P 500 market risk premium; and second, a factor of 25% multiplied by the S&P 500 market risk premium. The zero-beta CAPM expression takes the form: $K = Rf + 0.75*\beta*(Rm-Rf) + 0.25*(Rm-Rf)$, where all the variables are the same as used in the standard CAPM. The zero-beta CAPM was developed to address the alleged infirmity that the traditional CAPM understated the required return of stocks with a beta below 1.

Q. What is the result of your zero-beta CAPM methodology?

A. As the following calculation illustrates, our zero-beta CAPM produces an 8.44% cost of equity.
Cost of equity = 2.55% ((0.75 *.61 (10.88%-2.55%)) + ((0.25 x (10.88%-2.55%)).

Q. What CAPM result did you use in the calculation of the proxy group cost of equity?

A. Consistent with Commission's precedent in prior rate proceedings, including its most recent decisions and orders in fully litigated cases, such as the recent NFG Rate Case, Case 16-G-0257, we have averaged the results of the traditional and zero-beta CAPM analyses to arrive at a CAPM result of 8.04%. The CAPM calculation is shown in Exhibit___(SFP-20).

Q. What role should the CAPM play in the cost of equity analysis?

A. In prior rate proceedings, like those to which we have previously referred to, the Commission has recognized that the CAPM presents a conceptual framework for calculating a reasonable estimate of a firm's cost of equity, however, because of the relative weaknesses of the CAPM in comparison with the DCF methodology, it is given a lesser weight than the DCF. The Commission has adopted a 33% weighting for the CAPM, compared to a 67% weighting of the DCF in the overall cost of equity analysis.

Q. What are some of the weaknesses of the CAPM

methodology?

A. The first shortcoming of the CAPM is its reliance on historical betas as a surrogate for forward-looking betas of companies. Historical estimates only reflect the past riskiness of an equity security that may not be representative of the future riskiness that is relevant to equity investors. The second weakness is that recent history has shown that volatility in betas can be dramatic. This may lead to estimating errors in the cost of equity. Finally, the traditional model may not adequately explain the variation in stock returns. To underscore the fact that the CAPM does not appear to adequately explain the variations in stock returns, Ambika Prasad Dash explained in his book entitled Security Analysis and Portfolio Management that "…empirical studies show that low beta stocks may offer higher returns than the model would predict." This excerpt is attached as Exhibit___(SFP-21).

Q. What cost of common equity did you calculate for your proxy group?

A. Using a 2/3 DCF and 1/3 CAPM weighting, the estimated cost of common equity is 8.16% rounded

up to 8.20%.

Q. How does the Panel's 8.20% recommendation compare to the Companies' currently authorized ROEs?

A. The Companies' are currently authorized an ROE of 9.0% dating back to the multi-year rate plans approved by the Commission in Cases 16-G-0058 and 16-G-0059 in December 2016. It is important to note, however, that when the Commission evaluates the reasonableness of a multi-year rate plan's ROE, it recognizes the necessity of a premium above the ROE produced using Staff's methodology in recognition of the increased financial and business risks that are inherent when a company is operating within the parameters of a multi-year rate plan. By agreeing not to file for new rates for three years, the Companies reduced their ability to adjust rates in response to changing conditions. Accordingly, locking in rates for an extended period somewhat diminished the Companies' flexibility and thereby increased their level of risk. Additionally, the forecasts of expense items in years two and three of a multi-year rate plan are inherently more inaccurate and thus expose a company to

additional operating risk. Finally, it is also worth noting that the interest rate environment when the Commission approved the current rate plans' 9.0% ROEs was modestly higher compared with today.

Q. Is there any reason a rational investor would expect the Commission to authorize an ROE in these proceedings anywhere close to the Companies' 9.65% requested ROE?

A. No. Rational investors are well aware of the Commission's preference for a formulaic approach to the cost of common equity and are also aware of the Commission's preference for establishing rates over a multi-year period. Accordingly, they likely expect that rates in this proceeding will be set over a multi-year period, and that the ROE, after accounting for the additional financial and business risk, would likely be similar to the 8.8% to 9.0% range of recent Joint Proposals approved by the Commission.

Q. What are the Joint Proposals that the Panel is referring to?

A. We are referring to the three most recent Joint Proposals approved by the Commission. The Joint

Proposal for Orange and Rockland, approved in March 2019, adopted an ROE of 9.0%. Further, the Joint Proposal for Central Hudson, approved in June 2018, adopted an ROE of 8.8%. Finally, the Joint Proposal for Niagara Mohawk, approved in March 2018, adopted an ROE of 9.0%.

Q. In the event that a Joint Proposal is not reached, and the Commission will be setting rates solely for the rate year, are there additional factors that it may want to consider with respect to the level of the Companies ROE?

A. Yes. We believe that there are two recent developments that could modestly and negatively impact investors return requirements. Specifically, once it has had sufficient time to analyze the issue, we believe the Commission should consider the impact of the Companies' moratorium on processing new natural gas service applicants in their respective service territories until the Northeast Supply Enhancement (NESE) gas pipeline receives its necessary permits. This development implies that the Companies will be potentially foregoing growth opportunities in the future. However, the

Commission's ROE authorization should also incorporate the impact of the likelihood that the Companies Revenue Decoupling Mechanism (RDM) will also be changed to a revenue per class reconciliation from the current revenue per customer model. In light of the curtailing of natural gas expansion in the Companies' service territories, the switch to a revenue per class RDM would offer the Companies' greater protection from a decline in sales.

Discussion of Companies' Witness Bulkley's ROE Approach

Q. Please summarize the approach followed by the Companies' witness Ms. Bulkley?

A. Ms. Bulkley began her analysis by applying her cost of equity models to two separate proxy groups. The first consisting of both electric utility holding companies and natural gas distribution holding companies, which she referred to as her "Combined Utility Proxy Group," and the second proxy group was derived solely from natural gas distribution holding companies, to which she referred to the "Natural

Gas Proxy Group". Ms. Bulkley developed three multi-stage DCF models utilizing the low, mean, and high earnings growth rate estimates published by Yahoo Finance, Zack's, and Value Line as shown in her Exhibit___(AEB-1). These DCF calculations resulted in a low, mean, and high ROE estimate for each company in her proxy group. Ms. Bulkley utilized both the traditional form of the CAPM and the zero-beta CAPM applied to both Bloomberg and Value Line betas that resulted in two sets of 12 ROEs for both her Combined Utility Proxy Group and Natural Gas Proxy Group. Ms. Bulkley then concluded that an appropriate ROE for the Companies ranges between 9.65% and 10.55%. In this proceeding, the Companies are seeking a 9.65% ROE for the rate year.

Q. What are your principal points of contention with Ms. Bulkley's analyses?

A. Overall, we are concerned with three primary aspects of her analysis: 1) the composition of both of the Companies' proxy groups, 2) the use of excessive growth rates in the Companies' DCF analyses, and 3) the use of several flawed key inputs employed in her CAPM analyses;

particularly Ms. Bulkley's market return estimate, and to a lesser extent the risk-free rate. Finally, we strongly disagree with Ms. Bulkley's premise that current market conditions warrant that the CAPM be accorded equal weighting with the DCF.

Criticism of The Proxy Groups Employed By Ms. Bulkley

Q. Please describe Ms. Bulkley's proxy groups.

A. Ms. Bulkley developed two proxy groups: one which she referred to as a Combined Utility Proxy Group, and the other which she referred to as a Natural Gas Proxy Group. The Natural Gas Proxy Group consists solely of the seven gas distribution companies that are also included in her Combined Utility Proxy Group. The Combined Utility Proxy Group consists of those seven gas companies and 13 additional electric distribution companies.

Q. How did Ms. Bulkley develop her Natural Gas Proxy Group?

A. Beginning with 10 companies classified as natural gas distribution companies by Value Line, she applied the following screening criteria: 1)

eliminated companies that are not covered by at least two utility industry equity analysts; 2) eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's; 3) eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source; 4) eliminated companies with less than 70 percent of total operating income derived from regulated utility operations; 5) eliminated companies that derive less than 50 percent of total operating income from regulated natural gas distribution operations; 6) eliminated companies known to be party to a merger, acquisition, or other transformational transaction; and 7) eliminated companies that have owned generation comprising greater than 60 percent of the company's MWh.

Q. Do you believe that Ms. Bulkley's use of a Natural Gas Proxy Group is appropriate?

A. No, the primary reason Ms. Bulkley's use of a Natural Gas Proxy Group is inappropriate because National Grid is viewed principally as an

electric utility holding company by investors. Also, even in instances where it is appropriate to look at the gas utility industry such as with companies like NFG, Corning and St. Lawrence Gas, a Natural Gas Proxy Group, when constructed using Staff's long-running selection criteria, would only produce a five company proxy group, and a proxy group consisting of only five companies does not provide an adequate sample size to produce reliable results. An inadequate sample size introduces the possibility of a single analyst error or the presence of an outlier to influence the cost of equity results as evidence in Ms. Bulkley's use of Northwest Natural Gas Company's growth rate shown on page 6 of her Exhibit__(AEB-1).

Q. Can you provide any research indicating what a reasonable size of a proxy group should be?

A. Yes. In the book "New Regulatory Finance" by Dr. Roger Morin, he suggests that a proxy group size of between 15 and 30 is reasonable to support a reliable cost of equity estimate.

Q. Can you provide an excerpt from the book to support your assertion that a proxy group size of

between 15 and 30 is reasonable?

A. Yes. In discussing cluster analysis to select a proxy group from a universe of pure-play utility companies, Dr. Morin states on pages 225 and 226 of his book that, "With cluster analysis, comparable companies are selected on the basis of 'closeness' to the targeted entity in terms of such predetermined risk variables such as credit ratings, after-tax interest coverage, equity ratio, total capital, and variability of operating income… A cluster size of 15 to 30 firms is reasonable. Once a cluster of comparable companies is identified for each firm or division, the average DCF and CAPM/ECAPM (Empirical Capital Asset Pricing Model) cost of equity estimates for the publicly traded companies are used as estimates of equity costs for the divisions." This excerpt is shown in Exhibit___(SFP-22). Even though the above-mentioned quote by Dr. Roger A. Morin is relative to cluster analysis, the principle is equally applicable to any proxy group selection irrespective of the selection criteria and methodology employed because unreasonable sample

size produces unreliable results.

Q. What were the selection criteria for Ms. Bulkley's second proxy group, the Combined Utility Proxy Group?

A. Ms. Bulkley began by selecting companies classified as electric utilities by Value Line, and then applied the following screening criteria: 1) eliminated companies that are not covered by at least two utility industry equity analysts; 2) eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's; 3) eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source; 4) eliminated companies that have owned generation comprising greater than 60 percent of the Company's MWh sales to ultimate customers; 5) eliminated companies with less than 70 percent of total operating income derived from regulated utility operations and 6) eliminated companies known to be party to a merger, acquisition, or other transformational transaction.

Q. What is the impact of Ms. Bulkley's selection criteria?

A. In the Companies' response to IR DPS-249, Ms. Bulkley indicated that, based on her six selection criteria, she eliminated 29 companies from the universe of 49 Electric and Natural Gas companies in constructing her Combined Utility Proxy Group.

Q. Do you have any concern with any of Ms. Bulkley's selection criteria?

A. Yes. We have three concerns regarding her selection criteria: 1) relating to net operating income, 2) analyst coverage, and 3) generation MWH sales.

Q. Please explain.

A. First, we find Ms. Bulkley's requirement that a company must have 70% of its net operating income from 'regulated utility operations' to be too restrictive. Operating income is the amount of revenue left over after accounting for all the expenses necessary to keep the business running. Her methodology would result in the elimination of companies whose risk characteristics are closely comparable to KEDNY/KEDLI for example,

NextEra Energy, Inc. was eliminated from her proxy group using the 70% operating income criteria. Moreover, using operating earnings instead of a revenue criterion for establishing the proxy group may allow companies into the group that contain substantially greater risk than a typical regulated utility. Revenues more accurately reflect the exposure to riskier competitive operations, because it is not uncommon for competitive ventures to be unprofitable at times. Accordingly, it is quite possible that using the 70% of operating income criteria could introduce companies into the proxy group that have a higher inherent risk profile than is proper for a regulated utility proxy group.

Q. Is there any other reason why it is inappropriate to use operating income as a selection criterion?

A. As illustrated in Exhibit___(SFP-23), the difference in accounting policy choices such as a difference in inventory valuation policy options and a difference in depreciation choices across companies could have a significant impact on reported operating income than on reported

revenues. For these reasons, revenue remains a superior criterion for ensuring that a utility proxy group properly reflects the amount of regulated business that most closely approximates the risk of the regulated utility being analyzed, and we recommend that Ms. Bulkley's use of operating income as a criterion should be rejected, and instead our 70% regulated revenue criterion be adopted.

Q. Please explain your second concern regarding her selection criteria?

A. Our second concern relates to the elimination of companies that are not covered by at least two utility industry equity analysts. Based upon her application of this criterion, Ms. Bulkley eliminated three companies: Otter Tail, MGE Energy, Inc. and El Paso Electric. It appears to us, however, that either Ms. Bulkley is not appropriately applying her own selection criterion, or she is unaware that these three companies are followed by more than one equity analyst. Specifically, Otter Tail is covered by Value Line Investment, Williams Capital Group, L.P, and Maxim Group, while El Paso Electric is

covered by Merrill Lynch, Mizuho Securities and Williams Capital Group as shown in our Exhibit___(SFP-24). Moreover, MGE Energy is covered by more than one equity analyst including CFRA Equity Research, Ford Equity Research Jefferson Research & Management with their latest reports published on July 19, 2019, August 9,2019, and August 9, 2019, respectively. Therefore, there is no reason to exclude Otter Tail, El Paso Electric, and MGE Energy from her proxy group on the basis of number of equity analysts.

Q. Please explain your final concern regarding her selection criteria?

A. We find Ms. Bulkley's limiting her proxy group to primarily transmission and distribution utilities (T&D) by excluding those having owned generation compromising greater than 60% of MWH sales to ultimate customers to be unreasonably restrictive. This restriction resulted in the elimination of eleven companies, that are all comparable in risk and thus suitable surrogates for the Companies. By drastically reducing the size of her proxy group, Ms. Bulkley has

undermined the reliability of her results. Thee eleven eliminated companies, which are all in our proxy group are: Ameren Corp., American Electric Power Co., Duke Energy, Entergy Corp., IDACORP, MGE Energy, NextEra Energy, OGE Energy, Pinnacle West Capital Corp., PNM Resources Inc. and Xcel Energy.

Criticism of Ms. Bulkley's DCF Methodology

Q. Please describe Ms. Bulkley's DCF methodology.

A. Ms. Bulkley uses a multi-stage DCF analysis based on a three-month average stock price and a range of near-term growth rate assumptions that produces a mean ROE of 9.20% for her Combined Utility Proxy Group and 9.04% for her Natural Gas Proxy Group. Specifically, she uses a three-stage DCF analysis that relied on equity analysts' earnings growth forecasts published by Value Line, Zacks and Yahoo Finance.

Q. Do you have any concerns with Ms. Bulkley's DCF analysis?

A. We have several concerns. Foremost among them, we find her third stage growth rate to be excessive and believe that she applies an

inappropriate internal rate of return (IRR) tool for calculating her ROE estimate. As we will explain shortly, we also find first-stage growth rates to be inferior to ours. Ms. Bulkley's use of an excessive third stage growth rate is the principal reason that her 9.20% DCF-based ROE recommendation is flawed. While the use of a three-stage DCF growth model is not necessarily problematic, our primary concern lies with her use of an excessive growth rate in her long run or third-stage growth rate, and to a lesser extent, some of the excessive growth rates she uses in the first stage of her analysis.

Q. Please elaborate why you believe that her first-stage growth rates are problematic.

A. Ms. Bulkley's model uses earnings growth rate estimates to determine short-term dividend growth rates, ignoring available dividend growth rates or forecasts. This is in direct contrast to the basic premise of the DCF which measures the present value of future dividends. In using earnings growth, Ms. Bulkley has simply assumed that dividend growth will match earnings growth but has presented no evidence why this

expectation is valid. Of greater concern is her use of a 25.5% earnings per share (EPS) growth rate for Northwest Natural Gas Company. We believe that a growth estimate of that magnitude is unreasonable and that she should have excluded it from her analysis. The impact of eliminating this 25.5% EPS growth rate would have reduced Ms. Bulkley's DCF results by nine basis points from 9.20% to 9.11%.

Q. Do you have any other concerns?

A. Yes. We also have a concern about her use of a historically derived real gross domestic product (GDP) rate, together with a forecasted inflation rate as the long-term growth rate of her proxy group. We will discuss this in detail later.

Q. Please describe how Ms. Bulkley derives her short-term dividend growth rates.

A. In the first stage of the three-stage DCF model, Ms. Bulkley takes the current annualized dividend for each company in her proxy group and escalates the dividend for a period of five years based on the average of the three-to-five-year earnings growth estimates reported by Yahoo Finance, Zacks and Value Line.

Q. Why is this approach problematic?
A. It is highly unlikely that investors would rely exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections. The 25-year old Harris study on page 59 of Ms. Bulkley's testimony purports that a "growing body of knowledge" shows that analyst earnings forecasts are indeed reflected in stock prices. While we agree that all relevant information is incorporated into a company's stock price, the linear relationship of earnings to dividend growth that she assumes is unlikely. That is, payout ratios will change based upon many factors including individual company cash flow requirements, future market conditions as well as other factors. Investors will not just look at expected earnings but also factor in all relevant information when estimating their required return.
Q. Does Ms. Bulkley have any concern with the use of Value Line short-term dividend projections?
A. Yes. Ms. Bulkley, on page 61 of her testimony, claims that since Value Line's projections of

short-term dividend growth do not explicitly include growth in retained earnings, their use is improper as inputs to the DCF analysis. What she has failed to recognize is that, while in the long-term, dividend growth is constrained to the combination of retention growth and issuing stock above/below book value, in the short run, dividend policy can change and result in dividends to grow above or below retention growth. Consequently, her short-term dividend projections are a direct product of the average earnings growth estimates of three different publications, without any consideration being given to the growth rates effect on future dividend payouts. Our use of Value Line dividend growth projections recognizes the impact of payout policy while her sole use of short-term earnings forecasts does not.

Q. Does Ms. Bulkley have other criticisms of using only Value Line dividend growth rates?

A. Yes. Ms. Bulkley claims Value Line's dividend projections are the growth expectations of a single analyst and attempts to discredit utilization of Value Line by claiming the

publication is based on a single analyst's viewpoint. While Value Line assigns a lead analyst in the electric utility industry who is then credited for each company's report, each report that Value Line develops goes through a continual evaluation and quality control process where 'multiple analysts' review the reports generated before they are ever posted. The Value Line document describing this process is included in Exhibit___(SFP-25).

Q. Is there another advantage to the continued use of Value Line projections in your analysis?

A. Yes. Staff and the Commission have reasonably relied on Value Line for many years. This methodology is generally understood by the investment community and lends a degree of predictability to the rate setting process in New York State. While this consistency does not help to identify the return that equity investors currently require, it is important in the sense that it provides predictability in the earnings level that investors in New York utilities can expect. This is particularly important to the major credit rating agencies, who view

unpredictability as an additional risk.

Q. You previously stated that Ms. Bulkley estimated long-term dividend growth based upon expected GDP growth. How did Ms. Bulkley calculate expected GDP growth?

A. Rather than using estimates of future GDP growth, Ms. Bulkley utilized a 3.22% historical average growth rate in real GDP for the period 1929 through 2018 based on data from the Bureau of Economic Analysis. Ms. Bulkley then calculated a 5.56% forecasted nominal GDP rate by taking this historical average growth rate, together with her expected average inflation rate of 2.27%. The 2.27% expected average inflation rate was calculated based upon an average of 2.2% expected inflation published by the Blue Chip Financial Forecasts, and 2.31% and 2.29% she derived using data from the Energy Information Administration (EIA). The EIA expected inflation was based on the compound annual Consumer Price Index (CPI) growth rate and the compound annual growth rate of the CPI for all urban consumers and the compound annual GDP Price Index. The derivation of her 5.56% long-term GDP growth rate is shown

on page 1 of her Exhibit__(AEB-2).

Q. Do you agree with Ms. Bulkley's 5.56% long-term growth rate estimate?

A. No. While we do not necessarily find it unreasonable to assume that long-run growth for utilities would approximate growth in the overall economy, what is unreasonable is the excessiveness of her estimate of long-term GDP growth. Specifically, we do not think the use of her historically-derived real GDP as the long-term growth rate is appropriate for reasons previously discussed, namely because historical growth rates are necessarily measured over a subjective time period, and in this instance, one that incorporates far different circumstances than exist today. Historical average growth rates by their very nature completely, and inappropriately ignore economic conditions today. In addition, the calculation presented by Ms. Bulkley does not accurately measure GDP growth. This inaccuracy occurs since the 2.27% expected inflation rate that she utilizes reflects expected price changes in the Consumer Price Index and in the CPI-Urban. The CPI measures

changes in the price level of a basket of consumer goods and services and, unlike the GDP deflator, does not measure inflation over the entire economy. Additionally, Ms. Bulkley's use of the real historical GDP growth rate from 1929 to 2018 is inappropriate because while historical averages provide insight into how past factors might have influenced past changes in GDP, they are poor indicators of future economic activity.

Q. Are there estimates of long-run growth in the economy that you believe provide a more reasonable estimate of long-run growth?

A. Yes, we are aware of two reasonable estimates of long-run growth. The first source is the Long-Range Consensus U.S. Economic Projections provided by Blue Chip Economic Indicators, which we previously discussed, and employed in ascertaining the reasonableness of our sustainable growth rate. Not only does this report project out into the future twice as far as any other reputable source of economic data, it also reflects the forward-looking consensus of approximately 50 of the financial community's most prominent economists. According to the

March 10, 2019 publication, which is the most recent and is also shown in our previously mentioned Exhibit___(SFP-18), the consensus long-run nominal GDP growth rate for the most distant period forecast, 2026-2030 is 4.0%, which includes both 2.1% real GDP growth and 1.9% expected inflation components. A reasonable investor would expect a similar long-term growth rate for an energy utility rather than the excessive 5.56% growth rate used by Ms. Bulkley.

Q. Please continue with your second source.

A. The second source as shown in our Exhibit___(SFP-SFP-26), is based on information and data from U.S. Energy Information Administration (EIA). On page 18 of its April 2019 Annual Energy Outlook, the EIA states that "The Reference, High Economic Growth, and Low Economic Growth cases illustrate three possible paths for U.S. economic growth. In the High Economic Growth case, average annual growth in real GDP is 2.4% from 2018 to 2050, compared with 1.9% in the Reference case. The Low Economic Growth case assumes a lower rate of annual growth in real GDP of 1.4%."

The EIA's optimistic long-run real GDP growth

rate of 2.4% suggests a high nominal GDP growth rate of about 4.5% (2.4% plus 2.1% inflation). Accordingly, in light of this information, Ms. Bulkley's use of an excessively high long-term growth rate of 5.56% totally undermines the reliability of her DCF results.

Q. Earlier, you indicated that you have a concern with Ms. Bulkley's use of the irregular internal rate of return in her DCF analysis. Please explain.

A. Our concern relates to Ms. Bulkley's inappropriate use of the extended version of the Internal Rate of Return (XIRR) in her DCF analyses versus the Commission preferred use of the regular IRR in our model. Her use of the XIRR overstates her mean DCF result of 9.20% for her Combined Utility Proxy group by approximately 15 bps and approximately 14 bps for her Natural Gas Proxy Group's mean DCF result of 9.04%.

Q. Please explain the XIRR and the IRR?

A. The XIRR is a Microsoft Excel function, or tool, used to calculate returns for a series of cash flows that are not regular. The IRR is suitable for regular cash flows that could be monthly,

quarterly, or annually. As an illustration, if an individual had six series of cash flows consisting of an initial investment of $1,000 made on June 30, 2014, and four dollar returns consisting of $200, $220, $400, and $300 on 2/28/2015, 1/28/2016, 2/15/2017 and 3/28/2018, respectively, this individual's investment return based on the irregular cash flow will be 5.0% compared to 4.3% using the regular IRR. Since dividend cash flow are typically paid on a regular basis, we do not believe that the XIRR is an appropriate tool for determining the DCF-based ROE.

Q. At various sections of her direct testimony, Ms. Bulkley, to some extent, cited the Recommended Decision (RD) issued in the Generic Finance Case (GFC) 91-M-0509 to support her assertion that the cost of equity methodologies that she employed are consistent with the goals of the Recommended Decision issued in the Generic Finance Proceeding. Did Ms. Bulkley provide any evidence indicating that the Generic Finance Case (both KEDNY and KEDLI were parties) either directly or indirectly relied on the use of the XIRR?

A. No. In response to IR DPS-895, Ms. Bulkley stated that she "…is unaware whether the Generic Finance Case RD relied on the IRR or XIRR formula."

Q. Panel, what is your opinion about whether the Generic Finance Case RD relied on the IRR or XIRR formula?

A. To understand whether the Generic Finance Case Recommended Decision relied on the IRR or XIRR, we reviewed and recast the DCF results for both the 34-member electric and the 11-member gas proxy groups in Appendix B of the GFC's "Return on Equity Consensus Document" prepared by the signatory members of the Electric and Gas Industry to which both KEDNY and KEDLI were signatory parties. For the 34-member electric proxy group, the GFC results shows a median and average DCF ROEs of 9.62% and 9.61%, respectively. Our recast using the regular IRR resulted in a 9.62% for both median and average ROE compared to 9.9% for both median and average ROE using the XIRR. For the 11-member gas proxy group, our recast of the GFC results shows a 10.70% and 10.40% average and median ROE,

respectively. Our recast using the regular IRR resulted in 10.71% and 10.40% average and median ROE, respectively compared to 11.0% and 10.67% average and median ROE using the XIRR. Our recast is shown in Exhibit___(SFP-27).

Q. Panel, what is your conclusion?

A. Our conclusion is that even though there was no specific discussion in the "Return on Equity Consensus Document" of the GFC concerning whether to use the regular IRR or XIRR, our analysis shows that the Generic Finance Case RD relied on the IRR rather than the XIRR for its DCF analysis at that time.

A. Has the Commission ever granted a cost of equity based upon a DCF analysis that discounted future dividends using XIRR?

A. No. Instead, both Staff and the Commission have relied on the regular IRR in DCF-based ROE calculations in both litigated and negotiated cases. Therefore, we recommend that the Commission reject Ms. Bulkley's use of the irregular internal rate of return and adopted Staff's position.

Q. Beginning on page 14 of her direct testimony, Ms.

Bulkley argues that due to the current market conditions, and in particular the Federal Reserve's monetary policies, the DCF model currently underestimates investors' required returns due to low dividend yields and high valuations on utility stocks, which she argues are not sustainable in light of her contention of investors' expectation of higher interest rates. How does the Panel respond to this assertion?

A. To gain better insight about trends in utility stock valuation and dividend yields, Staff compiled dividend yields, MBR and price-to-earning (P/E) ratio data from Merrill Lynch Quantitative Profiles from January 2005 to June 2019 as shown in Exhibit___(SFP-28). As the exhibit illustrates, the levels of utility valuation and dividend yields in the three years (2005-2007) preceding the 2008/09 credit and financial crisis are similar to or in some cases higher than the current levels. For instance, from 2005 to 2007, the average P/E ratio, MBR and dividend yields on annual basis were 16.49, 2.06 and 3.27%, respectively, compared to 16.23, 1.72, and 3.71%, respectively, for the post-financial

crisis period (2011 to June 2019). This trend suggests that low-interest rates as a result of the Federal Reserve monetary policies are not necessarily the only contributing factor to high valuation and low dividend yields in utility stocks. Therefore, Ms. Bulkley's belief that the current higher equity valuations for utility stocks have been driven by the Federal Reserve's monetary policies resulting in the DCF understating investors' required return is not grounded in any statistical analysis but appears to be based on a priori assumptions. What Ms. Bulkley is calling abnormal is really not all that different from the economic conditions that prevailed over several years before the financial crisis hit in 2008. Our interpretation is shared by one article that was recently published in the Wall Street Journal on July 8, 2019.

Q. Could you please cite the Wall Street Journal article.

A. Yes. In the article entitled, "Can Low Rates Explain High Stock Prices? Not So Fast", Mark Hulbert stated that "[t]oday's low interest rates justify above-average price/earnings ratios.

That's the theory, anyway. Unfortunately, that theory doesn't hold up when you look at it through the lens of history." This article is provided as Exhibit___(SFP-29).

Q. Please continue with your response.

A. Ms. Bulkley's assertion goes against the crux of the efficient market hypothesis (EMH) underlying the DCF model that posits that stock prices reflect collective judgment of all stock market participants and therefore, reflect all known factors influencing the price of a stock, including the Federal Reserve's monetary policies.

Q. Are you aware of any financial literature that has come out subsequent to the Financial Crisis that supports continuing adherence to the efficient market hypothesis?

A. Yes. In discussing the importance and interpretation of stock prices in his book entitled, "The Cost of Equity - A Practitioner's Guide (2010 Edition)", David C. Parcell, specifically stated on page 88 of his book that "[t]he market price of a firm's stock represents the collective judgement of all stock market

participants as to the value of a firm at a particular point in time. The stock price takes into consideration the participants' interpretation of all relevant factors, such as past present and future earnings, the risk of these earnings, dividend policy and other factors." This article is shown as Exhibit___(SFP-30).

Q. Is there any indication that the Federal Reserve is likely to increase interest rates in the near term as claimed by Ms. Bulkley?

A. We do not believe so because the market expectation is that the Federal Reserve is likely to cut or maintain the current level for the near term as shared by one article that was recently published in the Wall Street Journal on July 19, 2019, attached as Exhibit___(SFP-31). In discussing recent signals from Federal Reserve Chairman and other officials, the author, Nick Timiraos stated that "Fed Chairman Jerome Powell set the stage last week for the first interest rate cut in over a decade during congressional testimony, when he signaled concern about global growth and the risk of a more prolonged shortfall

in inflation from the Fed's 2% target. Those developments strengthened the case for a somewhat easier policy stance." The author continued by saying, "Federal Reserve officials signaled they are ready to lower interest rates by a quarter-percentage point later this month, while indicating the potential for additional reductions, despite the recent surge in market expectations of a half-point cut. Officials aren't prepared for bolder action now, according to the officials' recent public statements and interviews, as they weigh concerns about a slowdown in global growth, an increase in trade-policy uncertainty and a pullback in inflation." Accordingly, the Federal Reserve cut the interest rate on July 31, 2019, by 25 basis points, for the first time since December 2008.

Q. Panel, what is your overall conclusion?

A. Our overall conclusion is that Ms. Bulkley's use of historically-derived derived long-term growth rate, 25.5% earnings per share (EPS) growth rate for Northwest Natural Gas Company, irregular IRR and expectation of higher interest rate are inappropriate and should therefore, be rejected

by the Commission.

Criticism of Ms. Bulkley's CAPM Methodology

Q. Would you please summarize Ms. Bulkley's CAPM approaches?

A. Ms. Bulkley provided a total of 24 CAPM-derived ROE estimates for her Combined Utility and Natural Gas proxy groups using the same CAPM methodologies that we employ. Ms. Bulkley calculated 12 ROEs using the traditional CAPM methodology and 12 ROEs using the zero-beta CAPM methodology, using two different beta sources in combination with three different risk-free rate estimates for each version of the CAPM.

Q. Please explain how Ms. Buckley derived each of the three major inputs necessary for her CAPM methodologies: the risk-free rate, beta and the market risk premium.

A. As shown in her Exhibit__(AEB-4), Ms. Bulkley's traditional and zero-beta CAPM methodologies use a risk-free rate based on the current three-month average yield (December 2018 to February 2019 yields) on 30-year Treasury bonds (3.05%), the near-term projected 30-year Treasury yield for

the five forward-looking quarters (2Q-2019 through Q2-2020) (3.28%), and the long-term projected 30-year Treasury yield for the period 2020-2024 (3.90%). Similarly, Ms. Bulkley developed a market risk premium based on an estimated S&P 500 required market return of 13.77% and the subtraction of each of the three risk-free rates we discussed above. Ms. Bulkley utilized two different beta determinations within each of her CAPM methodologies. For her Combined Utility Proxy Group, Ms. Bulkley applied its 0.691 Bloomberg average beta and its 0.645 Value Line average beta. For her Natural Gas Proxy Group, she applied its 0.70 Bloomberg average beta and its 0.686 Value Line average beta. Given these respective inputs, Ms. Bulkley then developed six traditional CAPM estimates of the cost of common equity for KEDLI and KEDNY for both proxy groups, ranging from 9.96% to 10.81%, and six zero-beta estimates of the ROE for both proxy groups ranging from 10.92% to 11.55%. By essentially averaging all of these results, Ms. Bulkley's CAPM methodology produced a mean ROE estimate of 10.95% for her Natural Gas Proxy

Group and 10.72% for her Combined Utility Proxy Group.

Q. What concerns do you have with Ms. Bulkley's CAPM analyses?

A. We have three principal concerns regarding Ms. Bulkley's CAPM analysis: 1) relating to her derivation of a 13.77% market return, 2) her beta determinations, and 3) her risk-free rate listed in order of magnitude of their impact. Collectively, the impact of these infirmities is an overstatement in her cost of equity estimate of a little over 200 bps.

Q. Please explain your concern with the approach she used to determine her 13.77% required market return.

A. The primary flaw with her CAPM approach is that it relies entirely upon a constant growth DCF analysis of the S&P 500. Her use of an excessively high 13.77% market return versus our 10.98% market return is the main reason for the large differences in her CAPM results and ours. Quite simply, the basic assumption of this approach, that the Bloomberg earnings growth rate estimates formulated for the next three-to-five

years will last into perpetuity, is not reasonable. This is precisely why Staff relies on the ex-ante estimate of the required return of the S&P 500 provided by Merrill Lynch, in our previously mentioned Exhibit___(SFP-19). Merrill Lynch's multi-stage DCF-derived required return does not make this unrealistic assumption, but rather uses three specific stages with varying durations depending on the facts and circumstances of the company being examined. The unreasonableness of using a constant growth DCF calculation to estimate the required market return is perhaps best illustrated by considering the fact that 39 of the S&P 500 companies in Ms. Bulkley's market return calculation have near-term earnings growth estimates in excess of 20.0% and range as high as 96.36% (Exhibit__(AEB-1)). It is plainly unreasonable that investors would assume that those companies would be able to maintain those extraordinary growth rates forever.

Q. At various sections of her direct testimony, Ms. Bulkley endorses recent conclusions the FERC has made with respect to ROE to support the

reasonableness of her approach to ROE. For example, on page 69, line 6 to 8, she stated that "…FERC now relies on analysts' estimates of earnings growth in the short-term and a long-term GDP growth rate as the measure of growth in the second stage." Could Ms. Bulkley have also equally considered the FERC's proportionate weighting of short-term and long-term growth rate to check the reasonableness of her 13.77% derived market return?

A. Yes. Ms. Bulkley could have considered the FERC's proportionate weighting of short-term and long-term growth rate at two-thirds and one-third, respectively to check the reasonableness of her 13.77% derived market return.

Q. What would have been Ms. Bulkley's market return if she had used the FERC's weighted growth rate approach?

A. As shown in our Exhibit__(SFP-32), Ms. Bulkley's market return would have been 11.37%, which is 240 basis points lower than her 13.77% market return.

Q. Does Ms. Bulkley's market return calculation have any other flaws?

A. Yes. Many of the companies in the calculation of her excessive 13.77% market return are not paying any dividends. The DCF calculation is premised upon the present value of future cash flows, in this case dividends. If companies are not paying dividends, then the whole DCF calculation is flawed because a DCF cannot be performed and therefore her 13.77% market return should be rejected because of flaws in her calculation.

Q. What is the impact of her use of the excessive 13.77% market return?

A. For her Combined Utility Proxy Group, the impact is an increase of about 166 basis points.

Q. Please explain your second concern regarding the derivation of Ms. Bulkley's beta estimates.

A. The Commission has consistently utilized Value Line betas, and one of the principal reasons for doing so is Value Line's superiority over other sources like Bloomberg.

Q. Are there any independent sources that indicate that Value Line Beta is superior?

A. Yes. In their recent book entitled Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice, H. Kent Baker and

Gerald S. Martin also wrote about the superiority of Value Line betas over Bloomberg betas. They wrote that "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg, which uses the following formula:
adjusted beta = historic beta*(0.67)+0.33."
After plotting historic betas, Value Line adjusted betas, and Bloomberg adjusted betas, the authors concluded that "the [Exhibits] convey a key message for analysts: be wary of blindly using historic betas from individual company data. The tails of the distribution for historic betas contain significant measurement error. A clear advantage of Value Line betas (or those from other providers who do more than just historic regression) is that they reflect professional judgment and attention to detail."
This report is attached as Exhibit__(SFP-33).

Q. Has the Value Line beta accuracy been tested?

A. Yes. In her empirical research work entitled "Whose Beta is Best?", Ms. Diana Harrington investigated the accuracy of Value Line betas and those betas reported by Merrill Lynch, Wilshire

Associates, Rosenberg and Associates (now BARRA) and others. For the period studied, Value Line consistently outperformed all other services over the forecast horizon of one to four years for public utilities. The study also indicated that increased data points increase the beta accuracy. The report is attached as Exhibit__(SFP-34). We have not come across any empirical research work that shows that Bloomberg beta estimates are superior to Value Line betas.

Q. What is the impact of using the blended Value Line and Bloomberg betas?

A. For her Combined Utility Proxy Group, the impact is an overstatement of about 21 basis points.

Q. Please explain your third concern relating to Ms. Bulkley's use of the 30-year Treasury and future estimates of its rate as the appropriate risk-free rate.

A. Ms. Bulkley argues that the sole use of the yield on the 30-year Treasury is appropriate because, in her view, utility companies represent long-duration investments. However, it has long been Commission policy to rely on the average of the 10-year and 30-year Treasury yields to arrive at

the risk-free rate as investors have a range of investment horizons. Furthermore, as the Commission has maintained for many years, current rates are the best indicator of future rates as they are based upon the latest available information to investors. The rationale for this approach is well-established and reflects the reality that there are investors with intermediate-term as well as long-term investment horizons.

Q. Ms. Bulkley argues that the Commission's preferred approach is flawed because it does not address the expected lives of the Companies' assets, the equity duration of the utility industry, or what Morningstar suggests is "the time horizon of the chosen Treasury security is that it should match the time horizon of whatever is being valued," as stated on page 72. Do you agree with her arguments?

A. No, we do not. While Ms. Bulkley is correct that utility plant assets have very long lives, and we would agree that sound financing practices generally dictate these long-lived assets be financed with similarly long-lived securities,

her conclusion that all utility equity investors have an investment horizon of 30 years is unsubstantiated and erroneous. One needs to look no further than the long-term debt obligations supporting the Companies own rate bases to understand that investors' have different time horizons.

Q. Is Ms. Bulkley's use of Treasury yield projections appropriate for use in the CAPM?

A. No. Her use of interest rate projections should be rejected. As Staff and the Commission have maintained for many years, and as mentioned earlier, current rates are the best indicator of future rates as they are based on the latest available information to investors.

Q. What is the combined effect on Ms. Bulkley's CAPM results if she had used a blended yield of 10-year and 30-year Treasury yields similar to the Commission's preferred approach instead of her use of projected 30-year Treasury yields?

A. The impact is an increase of about 13 basis points.

Q. Panel, what is your overall conclusion relating to your concerns of Ms. Bulkley's excessive

13.77% market return, her beta determinations, and risk-free rate derivation?

A. Our overall conclusion is the impact of these infirmities is an overstatement in her cost of equity estimate of a little over 200 bps. Therefore, we recommend that the Commission reject Ms. Bulkley's 13.77% market return, her beta determinations, and risk-free rate derivation and instead accept Staff's appropriate CAPM inputs for the risk-free rate, market risk premium and beta determination.

Equal Weighting of DCF and CAPM Results

Q. On page 79 of her direct testimony, Ms. Bulkley argues that "…it is reasonable to apply equal weighting to the DCF and CAPM methods when determining the ROE for the Companies." Why does Ms. Bulkley argue that the CAPM methodology should be accorded an equal weighting with the DCF methodology?

A. Ms. Bulkley's explanation to justify her equal weighting is based on three main points: 1) the effect of the Federal Reserve monetary policies on market conditions and its corresponding impact

on utility stock valuations, 2) other utility regulatory bodies' recent response to the alleged impact of current capital market conditions on the DCF model, and 3) market expectations of higher interest rates. She then concluded that the DCF method is producing ROE results that are significantly lower than the results derived from her CAPM calculations, and therefore, it is reasonable to place less reliance on the DCF results (page 79 lines 8-10).

Q. Please what is your response?

A. First, as we pointed out earlier, the MBRs and P/E ratios over the past nine years or so, have actually generally been lower than they were in the three-year period preceding the financial crisis. The significance of that timeframe is that it was a time period well before the Federal Reserve had begun its Quantitative Easing, and yet we had MBRs even higher than they are today. Therefore, Ms. Bulkley's whole contention that utility stock prices are extraordinarily high specifically because of the Federal Reserve's monetary policies is misguided. Moreover, Ms. Bulkley would have us believe that investors

believe these "anomalous" conditions will significantly change in the not too distant future. Conditions that have persisted for nearly a decade should not be considered anomalous instead they should be viewed as the new normal. If the valuations for dividend-paying stocks have increased, it simply means that investors view dividend payments favorably and perceive less risk in an investment in companies with higher dividend payouts as compared to companies paying lower or no dividends. As such, lower risk should translate into lower return requirements and should be reflected in required ROE estimations. This would hold true in a DCF or any analysis quantifying required returns.

Q. Please elaborate on her second argument that other utility regulatory bodies have responded to the impact of the current market conditions on the DCF results.

A. Ms. Bulkley stated that other utility regulatory bodies like the Illinois Commerce Commission, the Pennsylvania Public Utility Commission, and the Missouri Public Service Commission have all "considered" the current market conditions impact

on DCF results. She also stated that the FERC, in particular, has determined that current capital market conditions have caused the DCF model to understate equity costs for regulated utilities at this time. She continued by saying that the FERC recently proposed a methodology that reflects the commission's current view that investors rely on multiple ROE estimation models and as a result the FERC proposed ROE methodology that includes equal weighting of the DCF, CAPM, Risk Premium and Expected Earnings.

Q. What is your response to her second argument?

A. Ms. Bulkley's characterization that other utility regulatory bodies have recognized the understatement of the DCF is misleading and shows selection bias. In any case, the opinion of another commission should have no bearing on a New York State proceeding. Furthermore, Staff is not compelled to abandon its approach, nor would we recommend to the Commission to deviate from its approved approach.

Q. Please respond to Ms. Bulkley's contention that because her CAPM results are significantly higher than the results of her DCF analyses, her DCF

results must be understated.

A. To begin with, we have already shown that her DCF results are not understated. However, because she has chosen to employ a methodology that yields such an egregiously high market return in her CAPM analysis, it is not surprising that her DCF analysis yields a lower ROE determination. However, such a finding would not be the case if her CAPM analysis employed reasonable assumptions as does ours. Under Staff's cost of equity model, the 8.10% CAPM result is currently lower than the 8.22% DCF result.

Q. Why should the CAPM be given less weight than the DCF in the weighting of the ROE methodologies?

A. As recognized by the Commission in prior rate proceedings, the CAPM presents a conceptual framework that provides a reasonable estimate of a firm's cost of equity, however, given some of the perpetual weaknesses of the CAPM discussed previously in our testimony, the Commission should continue its weighting of 33% compared to the DCF weighting of 67%.

Q. Has the Commission ever addressed the issue of equal weighting of the DCF and CAPM?

A. Yes. The Commission has consistently used the DCF as its primary methodology for determining ROE by applying a 2/3 weighting in the determination of the cost of equity. About a decade ago, some company witnesses in Cases 13-W-0539 & 13-W-0564 involving Suez Water Westchester, formerly known as United Water Westchester and New Rochelle, raised concerns about the weighting accorded the DCF and argued for equal weighting of the DCF and CAPM when the market-to-book ratio exceeds one or when the DCF was producing lower results than other methodologies the witnesses were advocating. In all the cases that we reviewed, the Commission repeatedly affirmed its preference for the two-thirds and one-third weighting of the DCF and CAPM, respectively. For example, we noted that the 2/3 DCF and 1/3 CAPM weighting was affirmed in 2008 by the Commission's Order in Case 07-E-0523 Consolidated Edison Company of New York, Inc.- Electric Rates, and in 2007 by the Commission's Order in Case 07-G-0141, National Fuel Gas Supply Co. – Gas Rates. There, the Commission stated, "[w]e also agree with Staff,

CPB, and Multiple Interveners that the Company (NFG) has not provided any compelling reasons to provide equal weight to the DCF and the CAPM methods... Accordingly, we will continue to use the two-thirds DCF Method and one-third CAPM method weighting in this case." Similarly, on pages 14 and 15 of the Commission's Order in Cases 06-E-1433 and 06-E-1547, Orange & Rockland Utilities, Inc. - Electric Rates, the Commission stated, "We will continue to accord two-thirds weight to the DCF result and one-third to the CAPM result as we have in past decisions." It is worthy of note that the Commission affirmed the proportionate weighting of the DCF and the CAPM when the utility stock valuations and dividend yields were similar to the current levels as discussed earlier. Therefore, we believe that Ms. Bulkley has not presented any new evidence that would compel the Commission to reject its long-standing policy of a two-third and one-third weighting of the DCF and CAPM and as such Ms. Bulkley's approach should be again rejected.

Impact of Correcting the Flaws in Ms. Bulkley's ROE Approach

Q. What is the combined effect on Ms. Bulkley's results if she had corrected the principal flaws in her DCF approach including 1) the use of the extended version of internal rate of return(XIRR) instead of the Commission preferred IRR version, 2) eliminated the excessive growth rate of Northwest Natural Gas Company, 3) properly relied on only Value Line betas, 4) used Merrill Lynch market returns, and 5) weighted her DCF results at two-thirds and CAPM results at one-third?

A. The combined effect of correcting the flaws in her DCF and CAPM approach including the inappropriate use of the extended version of IRR, rely on only the Value Line betas, and weight her DCF and CAPM results at two-thirds and one third, respectively is a reduction of 131 basis points from 9.71% to approximately 8.40%. The summary recalculation is shown in Exhibit___(SFP-35).

Regulatory Framework in New York

Q. On page 90 of her direct testimony, Ms. Bulkley stated that New York's regulatory framework has a

somewhat greater risk than the jurisdictions in which her proxy group companies provide service. Did Ms. Bulkley perform any qualitative analyses of state regulatory jurisdictions to support this assertion?

A. Ms. Bulkley performed three analyses as contained in her Exhibit___(AEB-9) and Exhibit___(AEB-10) to support her argument that the New York regulatory environment is relatively riskier. While Exhibit___(AEB-9) relates to KEDNY/KEDLI and Proxy Group Companies' Regulatory Research Associates (RRA) Rankings, Exhibit___(AEB-10) relates to S&P's assessment of regulatory credit supportiveness of the Companies and her proxy group.

Q. What is your opinion of Ms. Bulkley's statement that New York's regulatory framework has a somewhat greater risk than the jurisdictions in which her proxy group companies provide service?

A. Recent industry reports regarding the relative business and regulatory risks of New York's regulatory regime appear to find otherwise. For example, in May 2017, Regulatory Research Associates, a business unit of S&P, published the

results of its comprehensive audit of regulatory rankings in a report entitled, "State Regulatory Evaluations: Regulatory Climate for Energy Utilities Including an Overview of RRA's ranking process". Contrary to Ms. Bulkley's contention of higher business risk in New York, RRA viewed regulatory risk in the State as decreasing, as evidenced by its upgrading of New York's ranking from "Average/2" to "Average/1". This report is attached as Exhibit___(SFP-36).

The table on page 3 of that report only ranks nine, out of 53, national regulatory jurisdictions with higher evaluations relative to New York, which implies that RRA finds New York to have a more-favorable-than average regulatory environment.

Q. What aspects of New York's regulatory environment has RRA highlighted in its assessment?

A. In regards to New York's authorized electric and gas ROEs, RRA states that the Commission, "…in rate cases decided in recent years, has authorized electric and gas ROEs that are lower than the nationwide industry averages, for the most part, these decisions were based on multi-

year settlements that incorporated increasing rate bases over the term of the plans, revenue decoupling mechanisms and deferral accounting for increases in such items as net plant, pension expense, and labor costs. Additionally, other factors in the rate-setting process, including the incorporation of fully forecasted test periods improve the utilities' opportunity to earn the authorized ROE". This is shown as Exhibit___(SFP-37).

Q. Have any of the major credit rating agencies commented on New York's regulatory environment in recent credit reports?

A. Yes. Moody's issued a credit opinion for National Grid USA on July 20, 2017, which is provided in Exhibit___(SFP-38). On page 3 of that report, Moody's stated that "[w]hile our view of regulation in the US has improved generally, we regard New York as one of the most creditor-friendly jurisdictions. Recent rate case settlements have allowed rates to increase and included de-risking provisions such as timely cost recovery, forward-looking test years for operating expenditure and revenue decoupling

(designed to provide stable fixed cost recovery). Collectively, these provisions have provided more transparency and will likely result in lower cash flow volatility going forwards. In addition, a number of utilities regulated by the [Commission] are required to maintain a certain financial profile either through explicit leverage restrictions (KeySpan Gas East Corporation, known as KEDLI, and Brooklyn Union Gas Company, known as KEDNY) or are at least required to maintain an investment-grade rating (KEDLI, KEDNY and Niagara Mohawk Power Corporation, NiMo, A2 stable). KEDLI, KEDNY and NiMo are also subject to a 'golden share' provision which limits the potential for financial distress at the parent negatively affecting individual subsidiaries."

Q. Does New York regulation incorporate other risk-reducing elements in addition to those cited by Moody's?

A. Yes, New York regulation incorporates many important elements that have a direct impact upon a utility's ability to achieve its authorized return on equity. For instance, New York uses a fully forecasted test year to determine revenue

requirement, which recognizes changes in revenues and expenses expected through the end of the rate year. The Commission also generally allows for a high level of expense reconciliation for such material cost items such as environmental remediation costs and pension and OPEB expenses. New York also uses purchase power adjustment clauses that allow full and timely recovery of this large and volatile cost element. Many other jurisdictions do not incorporate these mechanisms into their rate plans. For overall blanket-type protection, New York State, unlike most other states, allows utilities to request deferral accounting for material items actually incurred but not provided for in the base forecasts.

Q. Please discuss the impact of a fully forecasted rate year upon a utility's risk profile.

A. During periods of rising operating and maintenance costs, combined with the need for significant capital additions, we believe that New York State's methodology of a fully-forecasted rate year provides utilities with a much greater opportunity to earn their authorized return than jurisdictions that establish rates

based on an adjusted or unadjusted historic test year. In addition, New York State regulation allows utilities to revise their rate filings during the suspension period for known changes in cost rates, changes in estimates, and for wholesale revisions because of changed circumstances beyond the company's control. This significantly reduces risk and improves earnings.

Q. Do you believe that investors are aware of such risk reducing elements of NYS utility regulation?

A. Yes. The NYS regulatory environment was presented by Commission Chair in a message to RRA clients, dated October 29, 2014. We note two main points in the report. First, the Commission's reliance on fully forecasted test years provides NYS utilities with mechanisms whereby expenses, capital additions, and returns are forecast through the end of the year for which rates are being set. Therefore, earnings are not adversely affected by inflation, new programs, and plant additions. Second, New York utilities have a strong history of being able to achieve healthy returns largely because of New York's risk-reducing mechanisms such as fully

forecasted test years, revenue decoupling mechanisms, deferrals and true-up mechanisms for pension, other post-employment benefits, and property taxes. This report is shown in Exhibit___(SFP-39).

Q. How many of their existing true-up and deferral mechanisms have the Companies requested be continued in this proceeding?

A. The Companies have requested for continuation of all of their existing true-up and deferral mechanisms.

Q. Please comment on the Companies' requested deferral mechanisms in this proceeding.

A. Even though not all the requested true-ups have been blessed by Staff, we believe that some of KEDLI/KEDNY's revenues and costs would be reconciled and therefore there will be no substantial risk for investors. Here are a few examples. First, Staff Efficiency and Sustainability Panel recommends an energy efficiency (ETIP) reconciliation that aligns with the Commission's direction in the March 2018 ETIP. Second, the Staff Consumer Services Panel agrees with the companies' proposal to fully

reconcile credit card and debit card costs but with some modifications. Third, Staff witness for the New York Facilities System Surcharge (NYFS) agrees with the Companies' proposal to change the timing of their NYFS reconciliations. Fourth, the Staff Gas Supply Panel agrees with the Companies' proposal to use 100 percent of options as a financial hedge against volatility in gas cost because such change will provide an added level of diversification for mitigating gas cost volatility. If granted by the Commission, the resulting risk reduction would reduce the need for thicker common equity ratio and the equity return requirements.

Q. Does Ms. Bulkley take these deferral mechanisms into account in her ROE recommendation?

A. No. Ms. Bulkley did not take into account the various deferral mechanisms in her ROE recommendation.

Q. Does Ms. Bulkley's return on equity request accurately reflect the Companies' deferral requests?

A. No, Ms. Bulkley has not made an attempt to compare the extensive suite of true-up and

deferral mechanisms being requested by the Companies with such mechanisms in place for her surrogate group. Absent such an analysis, it is very likely that Ms. Bulkley has overestimated the cost of equity for the Companies.

Q. Have the Companies proposed any Earnings Adjustment Mechanisms (EAMs) as part of the REV initiative?

A. Yes. The Companies propose three categories of EAMs, totaling six separate metrics. The three categories consist of: System Efficiency, Energy Efficiency, and Carbon Reduction. KEDNY could earn up to 78 basis points at maximum attainment levels for calendar year (CY) 2020 and 63 basis points for CYs 2021, 2022, and 2023. KEDLI could earn up to 77 basis points for CY 2020, 62 basis points in 2021, 63 basis points in 2022 and 64 basis points in 2023. Staff is recommending EAMs relating to Peak Reduction and Incremental Energy Efficiency worth about 25 basis points for the rate year.

Q. If approved by the Commission, how might these EAMs impact the Companies?

A. Successful implementation of these new

initiatives would likely enable the Companies to enhance their earnings above and beyond their authorized ROE.

Q. Does Ms. Bulkley take into account the opportunities afforded by the EAMs in her ROE recommendation?

A. No. Ms. Bulkley did not take into account the additional potential new sources of earnings resulting from the EAMs in her ROE recommendation. Rather, she discussed only the number of customer service quality and gas safety performance metrics that could negatively impact the revenue adjustments of the Companies. Ms. Bulkley's failure to account for the EAMs incremental earnings is inconsistent with the Companies' presentation for a reasonable opportunity to earn extra basis points above and beyond the Companies requested 9.65% ROE. To be conservative, however, our recommended cost of equity and credit metrics analysis do not reflect any of the potential incremental earnings associated with the EAMs.

Conclusion

Q. What is your recommended cost of capital for the Companies?

A. As summarized in Exhibit___(SFP-2) referenced earlier, on a pre-tax basis, our recommended overall pre-tax rate of return for KEDNY is 7.58% and 7.54% for KEDLI based on a capital structure with a 48% common equity ratio and an 8.20% cost of equity. As we stated earlier, we dismiss Ms. Bulkley's recommended cost of equity range of 9.65% to 10.75% as too high. Therefore, we recommend that the Commission reject both Ms. Bulkley's recommended cost of equity range and the Companies' requested 9.65% cost of equity and instead adopt our recommended 8.20% cost of equity.

Q. Does this conclude your testimony at this time?

A. Yes, it does.